

STAR RX72N - rx_obstacle_detect Library
1.0.0

Generated by Doxygen 1.16.1

Chapter 1

Directory Hierarchy

1.1 Directories

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Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

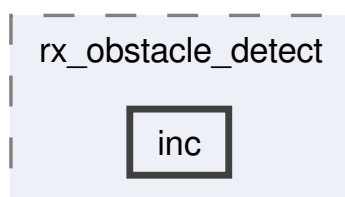
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Chapter 3

Directory Documentation

3.1 libs/rx_obstacle_detect/inc Directory Reference

Directory dependency graph for inc:

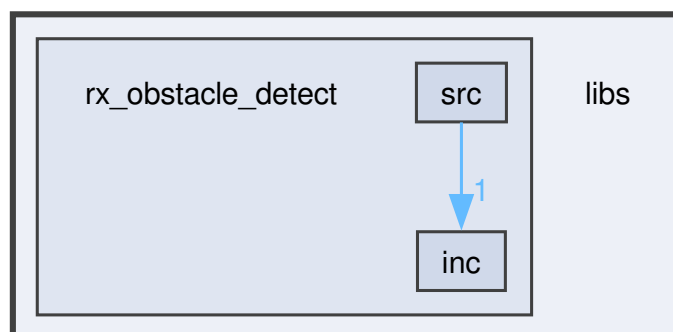


Files

- file [rx_obstacle_detect.h](#)
Safety-Critical Obstacle Detection System with Emergency Motor Stop.

3.2 libs Directory Reference

Directory dependency graph for libs:

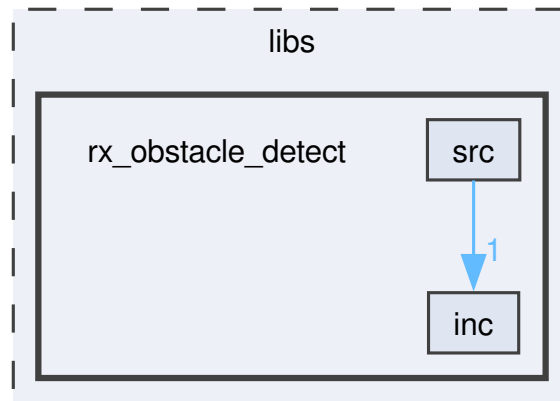


Directories

- directory [rx_obstacle_detect](#)

3.3 libs/rx_obstacle_detect Directory Reference

Directory dependency graph for rx_obstacle_detect:

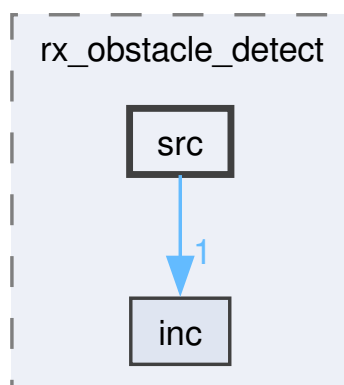


Directories

- directory [inc](#)
- directory [src](#)

3.4 libs/rx_obstacle_detect/src Directory Reference

Directory dependency graph for src:



Files

- file [rx_obstacle_detect.c](#)
Safety-Critical Obstacle Detection Implementation with ThreadX Task.

Chapter 4

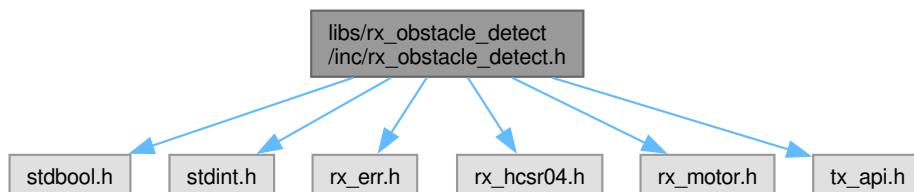
File Documentation

4.1 libs/rx_obstacle_detect/inc/rx_obstacle_detect.h File Reference

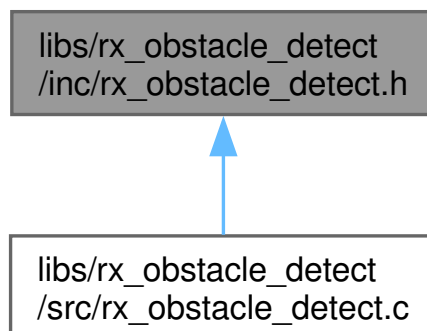
Safety-Critical Obstacle Detection System with Emergency Motor Stop.

```
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"
#include "rx_hcsr04.h"
#include "rx_motor.h"
#include "tx_api.h"
```

Include dependency graph for rx_obstacle_detect.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_obstacle_detect_config_t](#)
Obstacle detection system configuration parameters.
- struct [rx_obstacle_detect_t](#)
Obstacle detection handle.

Typedefs

- typedef void(* [rx_obstacle_detect_callback_t](#)) (bool obstacle_detected, uint8_t sensor_idx, float distance_cm, void *user_data)
User callback function for obstacle detection state change events.

Enumerations

- enum [rx_obstacle_detect_constants_t](#) : uint16_t { [k_obstacle_detect_max_sensors](#) = 8 , [k_obstacle_detect_max_motors](#) = 4 , [k_obstacle_detect_task_stack_size](#) = 2048 , [k_obstacle_detect_task_priority](#) = 8 }
Obstacle detection system configuration constants.
- enum [rx_obstacle_detect_state_t](#) : uint8_t { [k_obstacle_detect_state_stopped](#) = 0 , [k_obstacle_detect_state_running](#) = 1 , [k_obstacle_detect_state_obstacle](#) = 2 }
Obstacle detection system operational states.

Functions

- rx_err_t [rx_obstacle_detect_init](#) ([rx_obstacle_detect_t](#) *handle, const [rx_obstacle_detect_config_t](#) *config)
Initialize obstacle detection system with sensors and motors.
- rx_err_t [rx_obstacle_detect_deinit](#) ([rx_obstacle_detect_t](#) *handle)
Deinitialize obstacle detection system.
- rx_err_t [rx_obstacle_detect_start](#) ([rx_obstacle_detect_t](#) *handle)
Start obstacle detection monitoring.
- rx_err_t [rx_obstacle_detect_stop](#) ([rx_obstacle_detect_t](#) *handle)
Stop obstacle detection monitoring.
- rx_err_t [rx_obstacle_detect_clear_obstacle](#) ([rx_obstacle_detect_t](#) *handle)
Resume motors after obstacle cleared.
- rx_err_t [rx_obstacle_detect_get_state](#) (const [rx_obstacle_detect_t](#) *handle, [rx_obstacle_detect_state_t](#) *out_state)
Get current detection state.
- bool [rx_obstacle_detect_is_obstacle_detected](#) (const [rx_obstacle_detect_t](#) *handle)
Check if obstacle is currently detected.
- rx_err_t [rx_obstacle_detect_get_stats](#) (const [rx_obstacle_detect_t](#) *handle, uint32_t *out_total_polls, uint32_t *out_obstacle_events, uint32_t *out_false_positives)
Get obstacle detection statistics.
- rx_err_t [rx_obstacle_detect_reset_stats](#) ([rx_obstacle_detect_t](#) *handle)
Reset statistics counters.

4.1.1 Detailed Description

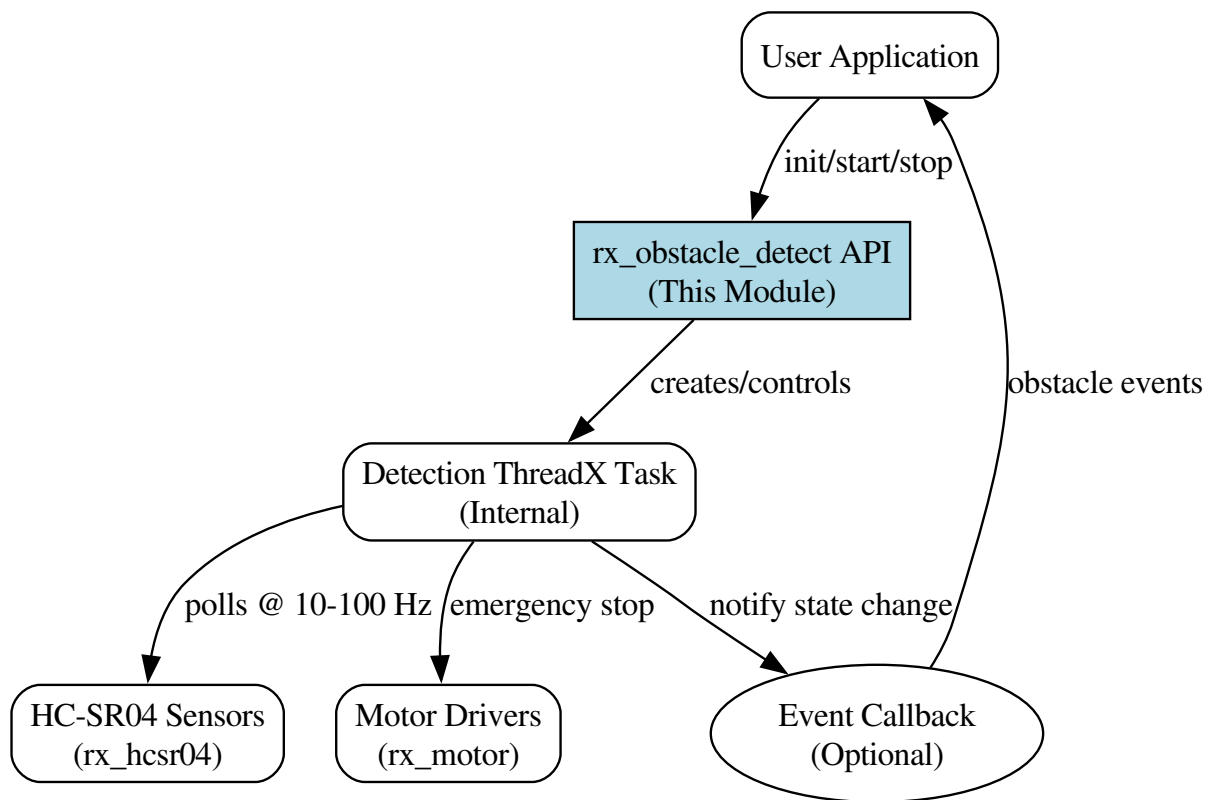
Safety-Critical Obstacle Detection System with Emergency Motor Stop.

Provides a ThreadX-based real-time obstacle detection system using HC-SR04 ultrasonic sensors with immediate emergency stop capability. This module continuously monitors configured sensors and automatically stops all motors when obstacles are detected within a configurable safety threshold.

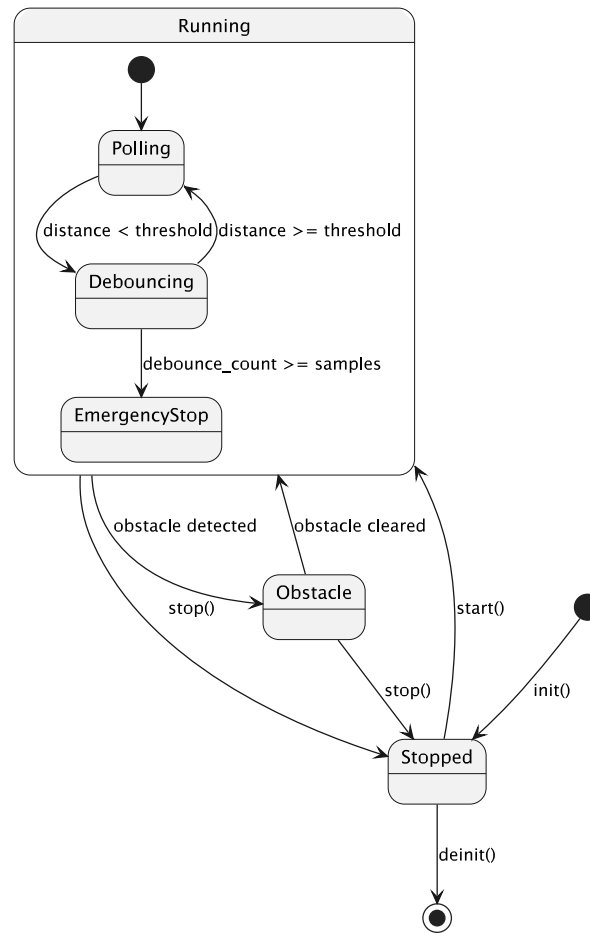
System Overview:

This is a simplified safety system focused on collision avoidance through emergency motor stop, not full Dynamic Window Approach (DWA) path planning. The module runs autonomously in a ThreadX task, polling HC-SR04 sensors at a configurable rate and triggering emergency stops when obstacles breach the safety perimeter.

Architecture:



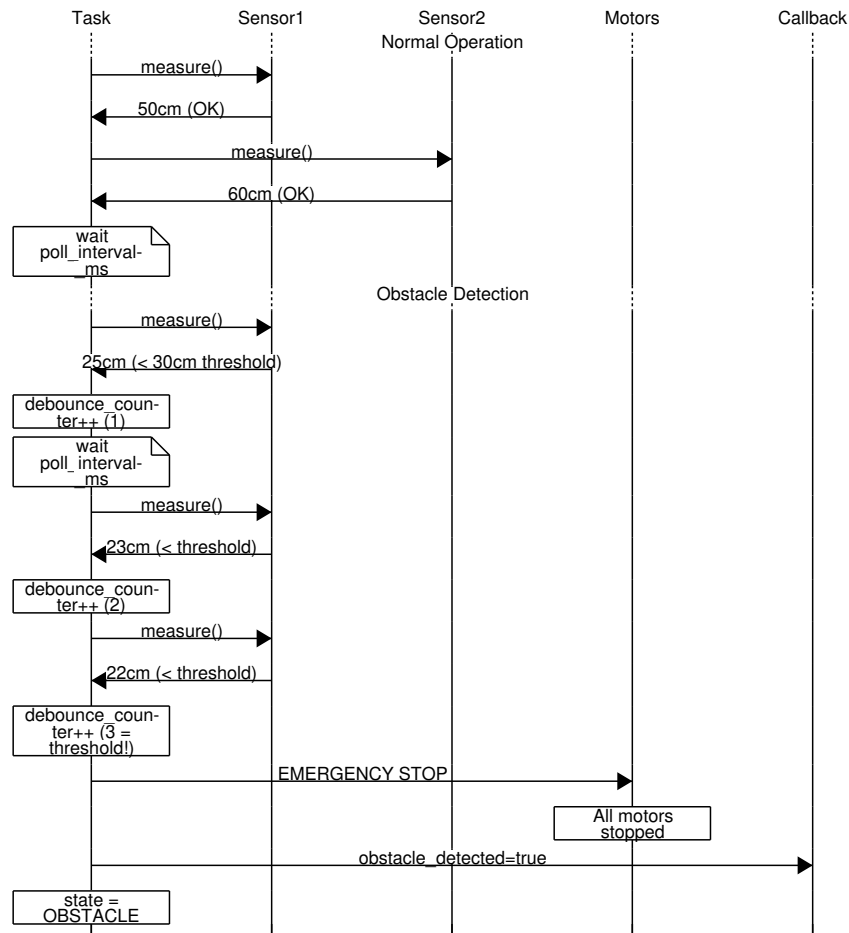
State Machine:



Features:

Feature	Description
Multi-Sensor Support	Up to 8 HC-SR04 sensors (front, sides, back)
Configurable Threshold	Stop distance 2-400cm (typical: 30cm)
Debouncing	1-10 consecutive readings to confirm (typical: 3)
Polling Rate	10-1000ms interval (typical: 20ms = 50Hz)
ThreadX Integration	Autonomous operation in dedicated task
Event Callbacks	Optional notification on state changes
Statistics	Track polls, events, false positives
Emergency Stop	Immediate motor shutdown on detection

Detection Logic:



Performance Characteristics:

Parameter	Value	Notes
Detection Latency	poll_interval_ms x debounce_samples	e.g., 20ms x 3 = 60ms worst-case
CPU Usage	~5% @ 50Hz, 4 sensors	RX72N @ 240 MHz
Memory (Static)	~2.5 KB	Stack (2048) + handle struct (512)
Memory (Dynamic)	0	Zero heap allocation
Thread Priority	8 (configurable)	Higher = more responsive

Safety Considerations:

Warning

This is a last-resort safety system, NOT primary collision avoidance
 Detection latency depends on polling rate and debounce settings
 Sensors have blind spots - physical robot design must account for this
 At maximum polling rate (100Hz, 1 debounce), minimum latency is 10ms
 At typical settings (50Hz, 3 debounce), worst-case latency is 60ms

Attention

Robot traveling at 1 m/s moves 6cm during worst-case 60ms detection

Always leave safety margin: $\text{threshold} > \text{max_travel_during_latency}$

For 1 m/s max speed, use $\text{threshold} \geq 10\text{cm}$ ($60\text{ms} \times 1\text{m/s} + \text{margin}$)

Hardware Requirements:

Resource	Requirement	Notes
CPU	RX72N @ 240 MHz	~12k CPU cycles per poll (4 sensors)
RAM	2.5 KB static	Task stack + handle structure
ThreadX	Required	Task creation and event flags
HC-SR04 Sensors	1-8 sensors	Each requires 2 GPIO pins
Motor Controllers	1-4 motors	Must support emergency stop

Module Dependencies:

- rx_hcsr04: HC-SR04 ultrasonic sensor driver
- rx_motor: Motor control with emergency stop capability
- rx_err: Error code definitions
- tx_api: ThreadX RTOS for task management
- rx_check: Assertion and validation macros
- rx_threadx_config: ThreadX configuration constants

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	[OK]	No goto, setjmp, recursion
2. Fixed loop bounds	[OK]	All loops use enum constants
3. No dynamic memory	[OK]	Zero malloc/free - static allocation only
4. Functions ≤ 60 lines	[OK]	Longest: internal_poll_sensors (54 lines)
5. Min 2 assertions/func	[OK]	All public APIs validate inputs
6. Smallest scope	[OK]	Variables declared at first use
7. Check return values	[OK]	All HC-SR04 and motor calls checked
8. Limit preprocessor	[OK]	Only C23 typed enums, no macros
9. Restrict pointers	[OK]	Max one level of dereferencing
10. Compiler warnings	[OK]	Compiles with -Wall -Wextra -Werror

SOLID Principles:

Single Responsibility (S):

- Module has ONE job: Monitor sensors and stop motors on obstacle detection
- Does NOT handle path planning, navigation, or motor control logic
- Does NOT configure sensors or motors - only uses them

Open/Closed (O):

- Extensible via configuration (threshold, debounce, polling rate)
- Open for extension via callback mechanism
- Closed for modification - core logic is stable

Dependency Inversion (D):

- Depends on rx_hcsr04 and rx_motor abstractions, not implementations
- Accepts sensor/motor handles via dependency injection (config)
- Callback mechanism allows user code to react without modifying module

See also

[rx_hcsr04.h](#) HC-SR04 ultrasonic sensor driver
[rx_motor.h](#) Motor control with emergency stop
[rx_obstacle_detect.c](#) Implementation file

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Date

2026-01-27

Copyright

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Version

1.0.0

Since

Version 1.0.0

Definition in file [rx_obstacle_detect.h](#).

4.1.2 Data Structure Documentation

4.1.2.1 struct rx_obstacle_detect_config_t

Obstacle detection system configuration parameters.

Configuration structure passed to `rx_obstacle_detect_init()` to set up the detection system. Defines which sensors and motors to use, detection thresholds, debouncing behavior, and optional event callbacks.

Configuration Guidelines:

Parameter	Typical Range	Recommended	Notes
sensor_count	1-8	4	Front coverage minimum
motor_count	1-4	4	All drive motors
threshold_cm	10-100	30	Depends on max speed
debounce_samples	1-5	3	Balance latency vs false positives
poll_interval_ms	10-100	20	50Hz good balance

Threshold Selection Formula:

Choose threshold based on maximum robot velocity and acceptable stopping distance:

$$\text{threshold}_{cm} \geq v_{max} \times t_{latency} + d_{stop} + \text{margin}$$

where:

- v_{max} = maximum robot velocity (cm/s)
- $t_{latency}$ = detection latency (s) = poll_interval x debounce_samples
- d_{stop} = braking distance (cm)
- margin = safety factor (typically 5-10cm)

Example Calculation:

- Max velocity: 100 cm/s (1 m/s)
- Poll interval: 20ms, Debounce: 3 -> Latency = 60ms = 0.06s
- Travel during latency: $100 \times 0.06 = 6\text{cm}$
- Braking distance: $\sim 10\text{cm}$ (depends on motor/PID)
- Margin: 10cm
- Threshold: $6 + 10 + 10 = 26\text{cm}$ -> use 30cm

Field Constraints:

Field	Constraint	Validation
sensors	Non-nullptr	Checked in init
sensor_count	[1, 8]	Checked in init
motors	Non-nullptr	Checked in init
motor_count	[1, 4]	Checked in init
threshold_cm	[2.0, 400.0]	HC-SR04 range
debounce_samples	[1, 10]	Checked in init
poll_interval_ms	[10, 1000]	Checked in init
callback	Can be NULL	Optional
user_data	Can be NULL	Optional

Usage Example:

```
// Setup sensors
rx_hcsr04_t sensor_front, sensor_left, sensor_right, sensor_rear;
rx_hcsr04_t* sensors[] = {&sensor_front, &sensor_left, &sensor_right, &sensor_rear};

// Setup motors
rx_motor_handle_t motor_fl, motor_fr, motor_rl, motor_rr;
rx_motor_handle_t* motors[] = {&motor_fl, &motor_fr, &motor_rl, &motor_rr};

// Configure obstacle detection
rx_obstacle_detect_config_t config = {
    .sensors = sensors,
    .sensor_count = 4,
    .motors = motors,
    .motor_count = 4,
    .detection_threshold_cm = 30.0f, // 30cm safety perimeter
    .debounce_samples = 3,           // 3 consecutive readings
    .poll_interval_ms = 20,          // 50Hz polling
    .callback = obstacle_event_handler,
    .user_data = &app_context
};

rx_obstacle_detect_t detector;
rx_err_t ret = rx_obstacle_detect_init(&detector, &config);
```

See also

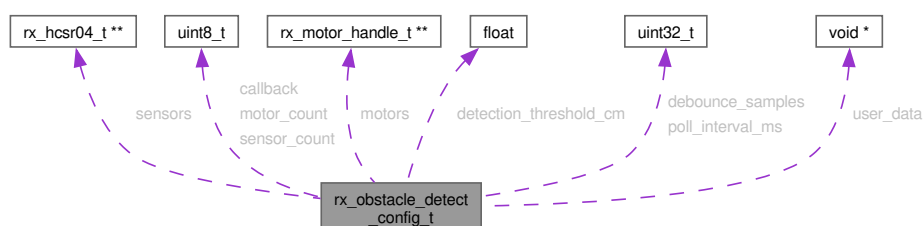
[rx_obstacle_detect_init\(\)](#) Uses this configuration
[internal_validate_config\(\)](#) Validates field constraints

Since

Version 1.0.0

Definition at line 671 of file [rx_obstacle_detect.h](#).

Collaboration diagram for rx_obstacle_detect_config_t:



Data Fields

rx_obstacle_detect_callback_t	callback	User-defined context pointer passed to callback. Arbitrary user data passed through to callback function. Typically points to application state structure. Set to NULL if not needed. Valid Values: Any pointer or nullptr Lifetime: Must remain valid while detection is running Typical Usage: • Pointer to application context struct • Pointer to telemetry buffer • Pointer to logging state Warning If non-NULL, must remain valid until deinit
---	----------	--

uint32_t	debounce_samples	Sensor polling interval in milliseconds. Time delay between sensor measurement cycles. Determines polling rate and CPU usage. Valid Range: [10, 1000 ms] Typical Values: • 10ms: 100Hz (very responsive, higher CPU usage) • 20ms: 50Hz (good balance, recommended) • 50ms: 20Hz (slower response, lower CPU usage) Unit: Milliseconds (ms) CPU Impact: • 100Hz (10ms): ~10% CPU @ 4 sensors • 50Hz (20ms): ~5% CPU @ 4 sensors • 20Hz (50ms): ~2% CPU @ 4 sensors Latency Impact: $\text{Worst-case detection latency} = \text{poll_interval_ms} \times \text{debounce_samples}$ Warning Lower intervals increase CPU load Higher intervals increase detection latency
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float	detection_threshold_cm	Number of consecutive sensor readings required to confirm obstacle. Debounce count to filter out transient sensor noise or spurious readings. Obstacle is only confirmed after this many consecutive readings below threshold. Valid Range: [1, 10] Typical Values: • 1: No debouncing (fastest response, more false positives) • 3: Good balance (recommended) • 5: Heavy debouncing (slower response, fewer false positives) Impact on Latency: $\text{detection_latency} = \text{poll_interval_ms} \times \text{debounce_samples}$ • 20ms x 3 = 60ms latency • 10ms x 3 = 30ms latency Warning Higher values increase detection latency
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uint8_t	motor_count	Obstacle detection threshold distance in centimeters. Minimum safe distance from obstacles. If any sensor measures distance below this threshold (after debouncing), emergency motor stop is triggered. Valid Range: [2.0, 400.0 cm (HC-SR04 specification)] Typical Values: • 10cm: Very close operation (indoor, slow speeds) • 30cm: General purpose (outdoor, moderate speeds) • 50cm: High-speed operation (fast robots) Unit: Centimeters (cm) Selection Guidelines: $\text{threshold} \geq (\text{max_speed_cm/s} \times \text{latency_s}) + \text{braking_distance_cm} + 10\text{cm margin}$ Warning Too small: Insufficient stopping time -> collisions Too large: Excessive false stops, limited mobility
rx_motor_handle_t **	motors	Number of motors in motors array. Count of valid motor pointers in the motors array. Valid Range: [1, k_obstacle_detect_max_motors (1-4)] Typical Value: 4 (four-wheel drive) Warning Must match actual motors array size

uint32_t	poll_interval_ms	Optional callback for obstacle state change events. Function pointer invoked when obstacles are detected or cleared. Set to NULL if no callback needed. Valid Values: Function pointer or nullptr Typical Usage: Logging, telemetry updates, UI notifications See also rx_obstacle_detect_callback_t Callback signature Warning Callback invoked from detection task - keep processing brief
uint8_t	sensor_count	Array of pointers to initialized motor control handles. Array of motor handles to stop when obstacle detected. Each motor must be pre-initialized via rx_motor_init() before passing to obstacle detection. Requirements: • Must be non-NULL • Must contain motor_count valid pointers • Each element must point to initialized rx_motor_handle_t • Array must remain valid for lifetime of detection system Typical Configuration: • 2 motors: Differential drive (left, right) • 4 motors: Four-wheel drive (FL, FR, RL, RR) Warning Motors must remain initialized while detection is running

rx_hcsr04_t **	sensors	Array of pointers to initialized HC-SR04 sensor handles. < Array of sensor handles to monitor for obstacles. Each sensor must be pre-initialized via rx_hcsr04_init() before passing to obstacle detection. Requirements: • Must be non-NULL • Must contain sensor_count valid pointers • Each element must point to initialized rx_hcsr04_t • Array must remain valid for lifetime of detection system Typical Configuration: • 4 sensors: Front, rear, left, right (90deg coverage) • 6 sensors: Frontx2, sidesx2, rearx2 (180deg coverage) • 8 sensors: Full 360deg coverage Warning Sensors must remain initialized while detection is running Number of sensors in sensors array Count of valid sensor pointers in the sensors array. Valid Range: [1, k_obstacle_detect_max_sensors (1-8)] Typical Value: 4 (front/rear/left/right) Warning Must match actual sensors array size
void *	user_data	

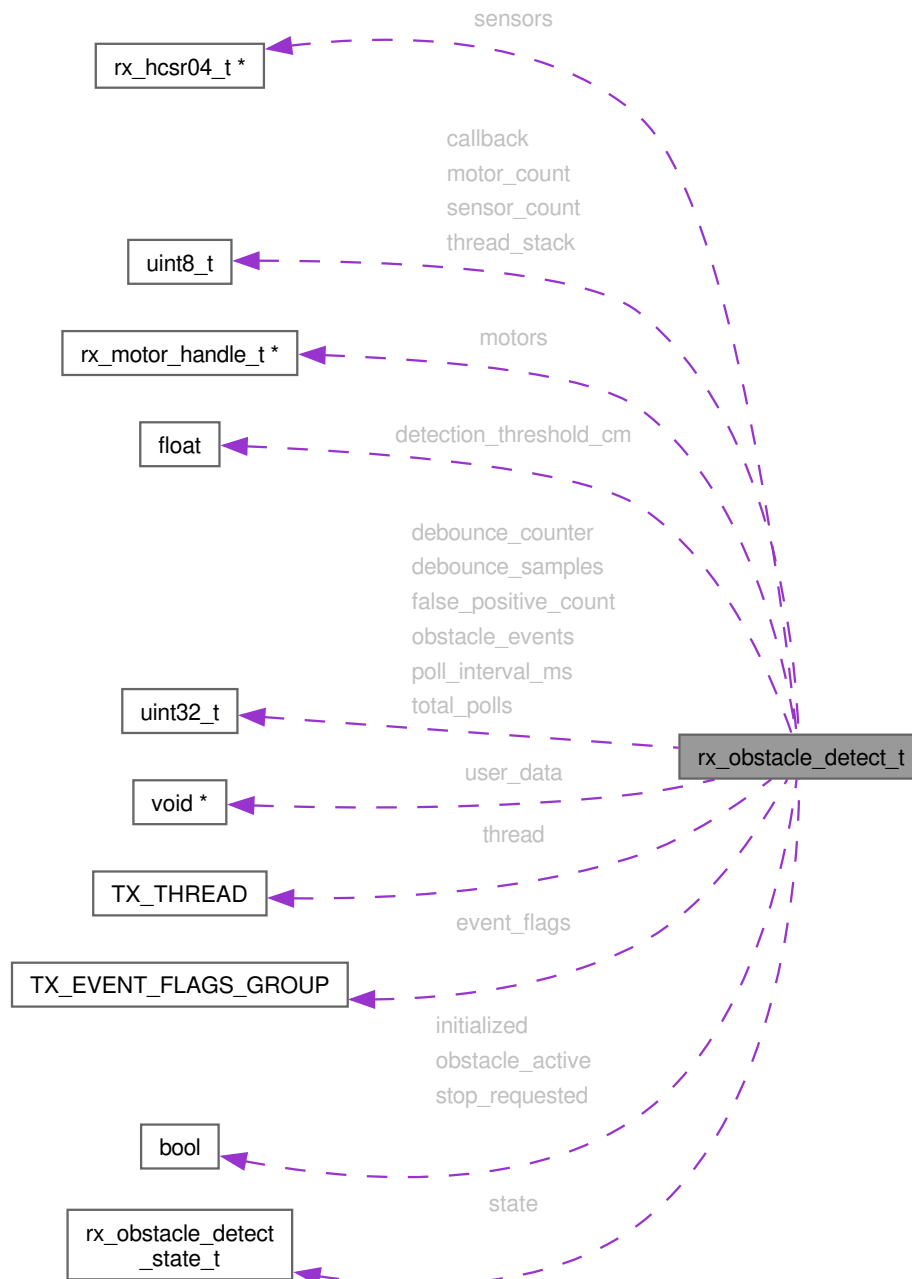
4.1.2.2 struct rx_obstacle_detect_t

Obstacle detection handle.

Maintains detection state and statistics. Caller allocates and passes to init. Do not modify fields directly after initialization.

Definition at line 847 of file [rx_obstacle_detect.h](#).

Collaboration diagram for rx_obstacle_detect_t:



Data Fields

rx_obstacle_detect_callback_t	callback	Event callback
uint32_t	debounce_counter[k_obstacle_detect_max_sensors]	Consecutive detections

uint32_t	debounce_samples	Debounce sample count
float	detection_threshold_cm	Stop distance threshold
TX_EVENT_FLAGS_GROUP	event_flags	Control event flags
uint32_t	false_positive_count	Debounced false positives
bool	initialized	True if initialized
uint8_t	motor_count	Number of motors
rx_motor_handle_t *	motors[k_obstacle_detect_max_motors]	Motor handles
bool	obstacle_active[k_obstacle_detect_max_sensors]	Per-sensor obstacle state
uint32_t	obstacle_events	Total obstacle detection events
uint32_t	poll_interval_ms	Polling interval
uint8_t	sensor_count	Number of sensors
rx_hcsr04_t *	sensors[k_obstacle_detect_max_sensors]	Sensor handles
rx_obstacle_detect_state_t	state	Current detection state
bool	stop_requested	True if stop requested
TX_THREAD	thread	Detection task thread
uint8_t	thread_stack[k_obstacle_detect_task_stack_size]	Task stack

uint32_t	total_polls	Total sensor polls performed
void *	user_data	User context

4.1.3 Typedef Documentation

4.1.3.1 rx_obstacle_detect_callback_t

```
typedef void(* rx_obstacle_detect_callback_t) (bool obstacle_detected, uint8_t sensor_idx, float distance_cm, void *user_data)
```

User callback function for obstacle detection state change events.

Optional callback invoked by the detection task when obstacle state changes:

- Obstacle detected (distance < threshold after debouncing)
- Obstacle cleared (distance >= threshold)

The callback is invoked from the ThreadX detection task context. Keep processing brief to avoid delaying sensor polling. Do NOT perform blocking operations or lengthy computations in the callback.

Callback Execution Context:

- Called from: ThreadX obstacle detection task
- Task priority: 8 (k_obstacle_detect_task_priority)
- Stack: Detection task stack (2048 bytes)
- Interrupt level: Task level (not ISR)

Typical Use Cases:

- Log obstacle events to console or storage
- Update UI/telemetry with obstacle status
- Trigger additional safety responses
- Notify higher-level navigation system
- Record statistics or diagnostics

Parameters

in	obstacle_detected	Obstacle state change direction • true: Obstacle detected (entering OBSTACLE state) • false: Obstacle cleared (returning to RUNNING state) • Type: bool
in	sensor_idx	Index of sensor that triggered this event • Range: [0, sensor_count-1] • Type: uint8_t • For detected events: Sensor that first exceeded debounce threshold • For cleared events: Last sensor that cleared • Maps to sensors[] array in rx_obstacle_detect_config_t
in	distance_cm	Measured distance in centimeters at time of event • Range: [2.0, 400.0] cm (HC-SR04 valid range) • Type: float • Unit: centimeters • For detected events: Final distance that triggered (< threshold) • For cleared events: Distance when sensor cleared (>= threshold) • May be slightly different from threshold due to polling intervals
in	user_data	User-provided context pointer from configuration • Type: void* (caller-defined) • Passed through from rx_obstacle_detect_config_t.user_data • Can be NULL if no context needed • Typical usage: Pointer to application state structure

Precondition

Callback is called from ThreadX task context (not ISR)
 sensor_idx < sensor_count (validated before callback)
 distance_cm within HC-SR04 valid range

Postcondition

Callback should return promptly (< 1ms recommended)
 Must not call blocking functions (sleep, pend, etc.)
 Must not call obstacle detection APIs (would deadlock)

Note

Thread Safety: Callback may be interrupted by higher-priority tasks

Re-entrancy: Will NOT be called recursively (single detection task)

Null Handling: If callback is nullptr in config, no callback invoked

Warning

DO NOT block or delay in callback - delays sensor polling

DO NOT call rx_obstacle_detect_* APIs from callback - causes deadlock

DO NOT perform heavy computation - run on detection task stack

Example - Logging Callback:

```
void obstacle_event_callback(bool obstacle_detected, uint8_t sensor_idx,
                             float distance_cm, void* user_data) {
    if (obstacle_detected) {
        printf("OBSTACLE! Sensor %u: %.1f cm\n", sensor_idx, distance_cm);
    } else {
        printf("Cleared. Sensor %u: %.1f cm\n", sensor_idx, distance_cm);
    }
}
```

Example - Telemetry Update:

```
typedef struct {
    telemetry_t* telem;
    uint32_t event_count;
} app_context_t;

void obstacle_telemetry_callback(bool obstacle_detected, uint8_t sensor_idx,
                                 float distance_cm, void* user_data) {
    app_context_t* ctx = (app_context_t*)user_data;
    ctx->event_count++;
    ctx->telem->obstacle_active = obstacle_detected;
    ctx->telem->min_distance_cm = distance_cm;
}
```

See also

[rx_obstacle_detect_config_t](#) Configuration includes callback and user_data

[rx_obstacle_detect_init\(\)](#) Registers callback during initialization

[internal_poll_sensors\(\)](#) Invokes callback on state changes

Since

Version 1.0.0

Definition at line 579 of file [rx_obstacle_detect.h](#).

4.1.4 Enumeration Type Documentation

4.1.4.1 rx_obstacle_detect_constants_t

enum [rx_obstacle_detect_constants_t](#) : uint16_t

Obstacle detection system configuration constants.

Defines limits and defaults for the obstacle detection system including maximum sensor/motor counts, ThreadX task configuration, and resource limits.

Value Rationale:

Constant	Value	Rationale
max_sensors	8	Typical robot: 4 front, 2 sides, 2 rear
max_motors	4	STAR robot has 4 drive motors
task_stack_size	2048 bytes	Sufficient for nested calls + local buffers
task_priority	8	Medium priority (0=highest, 31=lowest)

Task Priority Considerations:

- Priority 8 balances responsiveness with system load
- Higher priority (lower number) reduces detection latency
- Lower priority (higher number) reduces CPU contention
- Typical ThreadX priorities: 0-7 critical, 8-15 high, 16-23 normal, 24-31 low

See also

[rx_obstacle_detect_config_t](#) Uses these limits for validation

[rx_obstacle_detect_t](#) Allocates arrays sized by these constants

Since

Version 1.0.0

Enumerator

<code>k_obstacle_detect_max_sensors</code>	Maximum number of HC-SR04 sensors supported simultaneously. Maximum sensors that can be monitored in a single detection instance. Each sensor requires 2 GPIO pins (TRIG, ECHO) and ~64 bytes RAM. Value: 8 sensors Memory Impact: 8 x 64 bytes = 512 bytes (handle struct arrays) Typical Usage: 4-6 sensors for 360deg coverage Maximum number of motors controlled by emergency stop Maximum motors that will be stopped when obstacle detected. Each motor is an <code>rx_motor_handle_t</code> pointer in configuration. Value: 4 motors Memory Impact: 4 x sizeof(<code>rx_motor_handle_t</code>) = 16 bytes STAR Robot: Uses all 4 motors (front-left, front-right, rear-left, rear-right)
---	---

k_obstacle_detect_max_motors	Detection task stack size in bytes. Stack memory allocated for the ThreadX detection task. Must be sufficient for deepest call chain plus local variables and ThreadX overhead. Value: 2048 bytes (2 KB) Stack Usage Analysis: • ThreadX overhead: ~256 bytes • Task entry function: ~64 bytes • internal_poll_sensors(): ~128 bytes • rx_hcsr04_measure_blocking(): ~256 bytes (deepest call) • rx_motor_stop(): ~128 bytes • Margin: ~1216 bytes (safe margin for future changes) Rationale: 2048 bytes provides ~4x safety margin over measured usage (~512 bytes). ThreadX stack overflow detection will catch underruns during testing. Warning Insufficient stack causes undefined behavior (crashes, corruption)
------------------------------	--

k_obstacle_detect_task_stack_size	Detection task ThreadX priority level. ThreadX task priority for obstacle detection. Lower number = higher priority. Priority 8 provides good responsiveness without starving other tasks. Value: 8 (medium-high priority) Priority Levels in STAR System: • 0-3: Critical interrupt handlers • 4-7: Safety-critical tasks (e.g., motor control PID) • 8-11: High-priority tasks (obstacle detection) • 12-15: Normal application tasks • 16-31: Background/low-priority tasks Rationale: Priority 8 ensures: • Obstacle detection preempts normal tasks • Does not starve motor control (priority 5) • Typical 20ms poll interval leaves CPU time for other tasks Performance Impact: • @ 50Hz polling (20ms), task runs ~1ms/poll = 5% CPU @ 240 MHz • Leaves 95% CPU for other tasks See also tx_thread_create() ThreadX task creation uses this priority
k_obstacle_detect_task_priority	

Definition at line 237 of file [rx_obstacle_detect.h](#).

```

00237         : uint16_t {
00238     /**< @brief Maximum number of HC-SR04 sensors supported simultaneously
00239     *
00240     * @details
00241     * Maximum sensors that can be monitored in a single detection instance.
00242     * Each sensor requires 2 GPIO pins (TRIG, ECHO) and ~64 bytes RAM.
00243     *
00244     * @par Value: 8 sensors
00245     * @par Memory Impact: 8 x 64 bytes = 512 bytes (handle struct arrays)
00246     * @par Typical Usage: 4-6 sensors for 360deg coverage
00247     */
00248     k_obstacle_detect_max_sensors = 8,
00249
00250     /**< @brief Maximum number of motors controlled by emergency stop
00251     *
00252     * @details
00253     * Maximum motors that will be stopped when obstacle detected. Each motor
00254     * is an `rx_motor_handle_t` pointer in configuration.
00255     *
00256     * @par Value: 4 motors
00257     * @par Memory Impact: 4 x sizeof(rx_motor_handle_t*) = 16 bytes
00258     * @par STAR Robot: Uses all 4 motors (front-left, front-right, rear-left, rear-right)
00259     */

```



```

00260 k_obstacle_detect_max_motors = 4,
00261
00262 /**< @brief Detection task stack size in bytes
00263 *
00264 * @details
00265 * Stack memory allocated for the ThreadX detection task. Must be sufficient
00266 * for deepest call chain plus local variables and ThreadX overhead.
00267 *
00268 * @par Value: 2048 bytes (2 KB)
00269 *
00270 * @par Stack Usage Analysis:
00271 * - ThreadX overhead: ~256 bytes
00272 * - Task entry function: ~64 bytes
00273 * - internal_poll_sensors(): ~128 bytes
00274 * - rx_hcsr04_measure_blocking(): ~256 bytes (deepest call)
00275 * - rx_motor_stop(): ~128 bytes
00276 * - Margin: ~1216 bytes (safe margin for future changes)
00277 *
00278 * @par Rationale:
00279 * 2048 bytes provides ~4x safety margin over measured usage (~512 bytes).
00280 * ThreadX stack overflow detection will catch underruns during testing.
00281 *
00282 * @warning Insufficient stack causes undefined behavior (crashes, corruption)
00283 */
00284 k_obstacle_detect_task_stack_size = 2048,
00285
00286 /**< @brief Detection task ThreadX priority level
00287 *
00288 * @details
00289 * ThreadX task priority for obstacle detection. Lower number = higher priority.
00290 * Priority 8 provides good responsiveness without starving other tasks.
00291 *
00292 * @par Value: 8 (medium-high priority)
00293 *
00294 * @par Priority Levels in STAR System:
00295 * - 0-3: Critical interrupt handlers
00296 * - 4-7: Safety-critical tasks (e.g., motor control PID)
00297 * - **8-11: High-priority tasks (obstacle detection)**
00298 * - 12-15: Normal application tasks
00299 * - 16-31: Background/low-priority tasks
00300 *
00301 * @par Rationale:
00302 * Priority 8 ensures:
00303 * - Obstacle detection preempts normal tasks
00304 * - Does not starve motor control (priority 5)
00305 * - Typical 20ms poll interval leaves CPU time for other tasks
00306 *
00307 * @par Performance Impact:
00308 * - @ 50Hz polling (20ms), task runs ~1ms/poll = 5% CPU @ 240 MHz
00309 * - Leaves 95% CPU for other tasks
00310 *
00311 * @see tx_thread_create() ThreadX task creation uses this priority
00312 */
00313 k_obstacle_detect_task_priority = 8,
00314 } rx_obstacle_detect_constants_t;

```

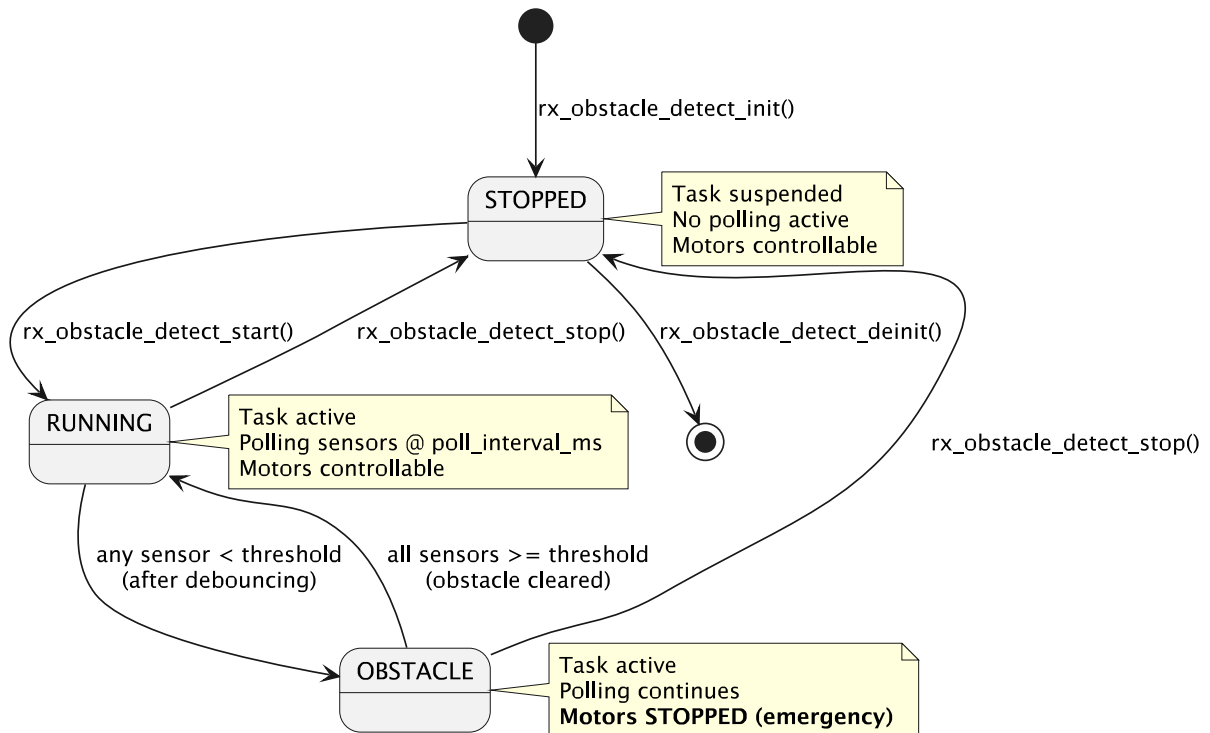
4.1.4.2 rx_obstacle_detect_state_t

```
enum rx_obstacle_detect_state_t : uint8_t
```

Obstacle detection system operational states.

Defines the three possible states of the obstacle detection system. State transitions are controlled by user commands (start/stop) and sensor measurements (obstacle detected/cleared).

State Transition Diagram:



State Transition Table:

Current State	Event	Next State	Action	Guard
STOPPED	start()	RUNNING	Resume ThreadX task	initialized == true
RUNNING	distance < threshold	OBSTACLE	Emergency stop motors	debounce_counter >= samples
RUNNING	stop()	STOPPED	Suspend task	none
OBSTACLE	distance >= threshold	RUNNING	(motors remain stopped)	all sensors clear
OBSTACLE	stop()	STOPPED	Suspend task	none
STOPPED	clear_obstacle()	STOPPED	Clear debounce state	none (manual override)

State Descriptions:

- **STOPPED:** System initialized but not monitoring. ThreadX task suspended. Motors controllable.
- **RUNNING:** Active monitoring, no obstacles. Task polling sensors at configured rate. Motors controllable.
- **OBSTACLE:** Obstacle within threshold. Motors emergency-stopped. Monitoring continues for clearance.

See also

[rx_obstacle_detect_get_state\(\)](#) Query current state

[rx_obstacle_detect_start\(\)](#) Transition STOPPED -> RUNNING

[rx_obstacle_detect_stop\(\)](#) Transition to STOPPED

[internal_poll_sensors\(\)](#) Implements RUNNING -> OBSTACLE transition

Since

Version 1.0.0

Enumerator

k_obstacle_detect_state_stopped	Detection system stopped - not monitoring sensors. Initial state after rx_obstacle_detect_init(). ThreadX detection task is created but suspended. No sensor polling occurs. Motors are controllable by application. Entry Actions: • Suspend ThreadX detection task • Clear event flags Do Actions: • None (task suspended) Exit Actions: • None Valid Transitions: • STOPPED -> RUNNING: via rx_obstacle_detect_start() Value: 0 (default/initial state) Detection active - monitoring sensors, no obstacle detected Normal operational state. ThreadX task actively polling all configured sensors at poll_interval_ms rate. No obstacles within threshold distance. Motors controllable by application. Entry Actions: • Resume ThreadX detection task • Set event flags (start flag) • Reset debounce counters Do Actions: • Poll each sensor via rx_hcsr04_measure_blocking() • Check distance against threshold • Increment debounce counter if distance < threshold • Clear debounce counter if distance >= threshold • Sleep for poll_interval_ms between polls Exit Actions: • Suspend task (if stop requested) • OR emergency stop motors (if obstacle confirmed) Valid Transitions: • RUNNING -> OBSTACLE: When debounce_counter >= debounce_samples • RUNNING -> STOPPED: via rx_obstacle_detect_stop()
Generated by Doxygen	Value: 1

k_obstacle_detect_state_running	Obstacle detected - motors emergency stopped. Critical state indicating obstacle within safety threshold. All configured motors have been emergency-stopped. Monitoring continues to detect when obstacle clears. Motors cannot be controlled until obstacle clears or user calls rx_obstacle_detect_clear_obstacle(). Entry Actions: • Call internal_stop_all_motors() (emergency stop) • Invoke user callback with obstacle_detected=true • Increment obstacle_events counter Do Actions: • Continue polling sensors • Check if ALL sensors clear (distance >= threshold) • Monitor for user stop request Exit Actions: • Invoke user callback with obstacle_detected=false (if clearing) • OR suspend task (if stop requested) Valid Transitions: • OBSTACLE -> RUNNING: When all sensors >= threshold (automatic) • OBSTACLE -> STOPPED: via rx_obstacle_detect_stop() Value: 2 Warning Motors remain stopped until obstacle clears or manual override Attention Application must NOT attempt to control motors in this state
k_obstacle_detect_state_obstacle	

Definition at line 382 of file [rx_obstacle_detect.h](#).

```

00382         : uint8_t {
00383     /**< @brief Detection system stopped - not monitoring sensors
00384     *
00385     * @details
00386     * Initial state after `rx_obstacle_detect_init()`. ThreadX detection task
00387     * is created but suspended. No sensor polling occurs. Motors are controllable
00388     * by application.
00389     *
00390     * @par Entry Actions:
```

```

00391 * - Suspend ThreadX detection task
00392 * - Clear event flags
00393 *
00394 * @par Do Actions:
00395 * - None (task suspended)
00396 *
00397 * @par Exit Actions:
00398 * - None
00399 *
00400 * @par Valid Transitions:
00401 * - STOPPED -> RUNNING: via `rx_obstacle_detect_start()`
00402 *
00403 * @par Value: 0 (default/initial state)
00404 */
00405 k_obstacle_detect_state_stopped = 0,
00406
00407 /**< @brief Detection active - monitoring sensors, no obstacle detected
00408 *
00409 * @details
00410 * Normal operational state. ThreadX task actively polling all configured
00411 * sensors at `poll_interval_ms` rate. No obstacles within threshold distance.
00412 * Motors controllable by application.
00413 *
00414 * @par Entry Actions:
00415 * - Resume ThreadX detection task
00416 * - Set event flags (start flag)
00417 * - Reset debounce counters
00418 *
00419 * @par Do Actions:
00420 * - Poll each sensor via `rx_hcsr04_measure_blocking()`
00421 * - Check distance against threshold
00422 * - Increment debounce counter if distance < threshold
00423 * - Clear debounce counter if distance >= threshold
00424 * - Sleep for poll_interval_ms between polls
00425 *
00426 * @par Exit Actions:
00427 * - Suspend task (if stop requested)
00428 * - OR emergency stop motors (if obstacle confirmed)
00429 *
00430 * @par Valid Transitions:
00431 * - RUNNING -> OBSTACLE: When debounce_counter >= debounce_samples
00432 * - RUNNING -> STOPPED: via `rx_obstacle_detect_stop()`
00433 *
00434 * @par Value: 1
00435 */
00436 k_obstacle_detect_state_running = 1,
00437
00438 /**< @brief Obstacle detected - motors emergency stopped
00439 *
00440 * @details
00441 * Critical state indicating obstacle within safety threshold. All configured
00442 * motors have been emergency-stopped. Monitoring continues to detect when
00443 * obstacle clears. Motors cannot be controlled until obstacle clears or
00444 * user calls `rx_obstacle_detect_clear_obstacle()`.
00445 *
00446 * @par Entry Actions:
00447 * - Call `internal_stop_all_motors()` (emergency stop)
00448 * - Invoke user callback with obstacle_detected=true
00449 * - Increment obstacle_events counter
00450 *
00451 * @par Do Actions:
00452 * - Continue polling sensors
00453 * - Check if ALL sensors clear (distance >= threshold)
00454 * - Monitor for user stop request
00455 *
00456 * @par Exit Actions:
00457 * - Invoke user callback with obstacle_detected=false (if clearing)
00458 * - OR suspend task (if stop requested)
00459 *
00460 * @par Valid Transitions:
00461 * - OBSTACLE -> RUNNING: When all sensors >= threshold (automatic)
00462 * - OBSTACLE -> STOPPED: via `rx_obstacle_detect_stop()`
00463 *
00464 * @par Value: 2
00465 *
00466 * @warning Motors remain stopped until obstacle clears or manual override
00467 * @attention Application must NOT attempt to control motors in this state
00468 */
00469 k_obstacle_detect_state_obstacle = 2,
00470 } rx_obstacle_detect_state_t;

```

4.1.5 Function Documentation

4.1.5.1 rx_obstacle_detect_clear_obstacle()

```
rx_err_t rx_obstacle_detect_clear_obstacle (
    rx_obstacle_detect_t * handle) [nodiscard]
```

Resume motors after obstacle cleared.

Clears the obstacle state and allows motors to be controlled again. This is a manual override - the user must verify the obstacle is truly cleared before calling this function.

If detection is still running and obstacle is still present, motors will be stopped again on the next poll cycle.

Parameters

in,out	handle	Obstacle detection handle
--------	--------	---------------------------

Returns

k_rx_ok on success
 k_rx_err_null_ptr if handle is nullptr
 k_rx_err_invalid_state if not initialized

Resume motors after obstacle cleared.

Clears all debounce counters and obstacle_active flags for every configured sensor. If the system is currently in k_obstacle_detect_state_obstacle, it transitions back to k_obstacle_detect_state_running.

This function is used to acknowledge a detected obstacle and allow normal operation to resume. It does NOT restart a stopped detection system.

Algorithm steps:

1. Validate handle is non-null and initialized
2. Iterate over all handle->sensor_count sensors
3. For each sensor: zero debounce_counter[i] and clear obstacle_active[i]
4. If state is k_obstacle_detect_state_obstacle, set state to k_obstacle_detect_state_running

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
 Detection system should be running (though clearing while stopped is safe)

Postcondition

All per-sensor debounce_counter values are zero
 All per-sensor obstacle_active flags are false

Note

Not thread-safe; clearing obstacle state while the detection task is concurrently updating sensor state may cause a race; caller must synchronize

Warning

Does not restart detection if stopped; use [rx_obstacle_detect_start\(\)](#)

Example:

```
// After handling obstacle, clear to allow normal operation
if (rx_obstacle_detect_is_obstacle_detected(&detector)) {
    handle_obstacle_event();
    rx_obstacle_detect_clear_obstacle(&detector);
}
```

See also

[rx_obstacle_detect_is_obstacle_detected\(\)](#) Check for active obstacle

[rx_obstacle_detect_get_state\(\)](#) Query current state

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 693 of file [rx_obstacle_detect.c](#).

```
00694 {
00695     /* Validate inputs */
00696     if (handle == nullptr) {
00697         return k_rx_err_null_ptr;
00698     }
00699
00700     if (!handle->initialized) {
00701         return k_rx_err_invalid_state;
00702     }
00703
00704     /* Clear obstacle state */
00705     for (uint8_t i = 0; i < handle->sensor_count; i++) {
00706         handle->debounce_counter[i] = 0;
00707         handle->obstacle_active[i] = false;
00708     }
00709
00710     /* Update state if currently in obstacle state */
00711     if (handle->state == k_obstacle_detect_state_obstacle) {
00712         handle->state = k_obstacle_detect_state_running;
00713     }
00714
00715     return k_rx_ok;
00716 }
```

References [rx_obstacle_detect_t::debounce_counter](#), [rx_obstacle_detect_t::initialized](#), [k_obstacle_detect_state_obstacle](#), [k_obstacle_detect_state_running](#), [rx_obstacle_detect_t::obstacle_active](#), [rx_obstacle_detect_t::sensor_count](#), and [rx_obstacle_detect_t::state](#).

4.1.5.2 rx_obstacle_detect_deinit()

```
rx_err_t rx_obstacle_detect_deinit (
    rx_obstacle_detect_t * handle) [nodiscard]
```

Deinitialize obstacle detection system.

Stops detection task, releases ThreadX resources, and resets handle state. Does NOT deinitialize the sensor or motor handles (caller's responsibility).

Parameters

in,out	handle	Obstacle detection handle
--------	--------	---------------------------

Returns

k_rx_ok on success
 k_rx_err_null_ptr if handle is nullptr
 k_rx_err_invalid_state if not initialized

Deinitialize obstacle detection system.

Tears down a fully initialized obstacle detection handle by stopping active detection if running, terminating and deleting the ThreadX monitoring thread, deleting the event flags group, and clearing the initialized flag. After this call the handle may be reused with [rx_obstacle_detect_init\(\)](#).

Algorithm steps:

1. Validate handle is non-null and initialized
2. If detection is not stopped, call [rx_obstacle_detect_stop\(\)](#) (errors ignored)
3. Terminate the ThreadX detection thread (TX_THREAD_ERROR ignored if already done)
4. Delete the ThreadX detection thread (TX_DELETE_ERROR ignored if already deleted)
5. Delete the TX event flags group (TX_DELETE_ERROR ignored if already deleted)
6. Clear handle->initialized

Precondition

handle must be a valid non-null pointer
 handle must have been successfully initialized via [rx_obstacle_detect_init\(\)](#)

Postcondition

handle->initialized is false
 ThreadX thread and event flags group are deleted

Note

Not thread-safe; caller must ensure no concurrent access to handle
 Stop errors during deinit are silently ignored; the teardown continues regardless

Example:

```
rx_obstacle_detect_t detector;
rx_obstacle_detect_init(&detector, &config);
// ... use detector ...
rx_err_t err = rx_obstacle_detect_deinit(&detector);
if (err != k_rx_ok) {
    rx_log_error("app", "Failed to deinit obstacle detector");
}
```

See also

[rx_obstacle_detect_init\(\)](#) Initialize the handle before use
[rx_obstacle_detect_stop\(\)](#) Stop detection before deinit

Since

Version 1.0.0

NASA Power of 10 Compliance:

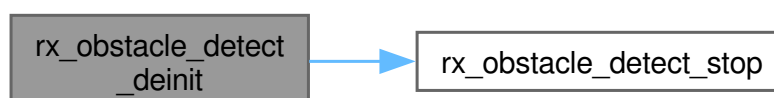
- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 460 of file [rx_obstacle_detect.c](#).

```
00461 {
00462     /* Validate inputs */
00463     if (handle == nullptr) {
00464         return k_rx_err_null_ptr;
00465     }
00466
00467     if (!handle->initialized) {
00468         return k_rx_err_invalid_state;
00469     }
00470
00471     /* Stop detection if running */
00472     if (handle->state != k_obstacle_detect_state_stopped) {
00473         rx_err_t stop_ret = rx_obstacle_detect_stop(handle);
00474         /* Note: Continue with deinit even if stop fails - we're tearing down anyway */
00475         /* Caller should check return value if stop failure is critical */
00476         (void)stop_ret;
00477     }
00478
00479     /* Wait for thread to terminate */
00480     UINT status = tx_thread_terminate(&handle->thread);
00481     const bool terminate_failed = (bool)((status != TX_SUCCESS) & (status != TX_THREAD_ERROR));
00482     if (terminate_failed) {
00483         return k_rx_err_rtos_error;
00484     }
00485
00486     /* Delete thread */
00487     status = tx_thread_delete(&handle->thread);
00488     const bool delete_failed = (bool)((status != TX_SUCCESS) & (status != TX_DELETE_ERROR));
00489     if (delete_failed) {
00490         return k_rx_err_rtos_error;
00491     }
00492
00493     /* Delete event flags */
00494     status = tx_event_flags_delete(&handle->event_flags);
00495     const bool evflags_failed = (bool)((status != TX_SUCCESS) & (status != TX_DELETE_ERROR));
00496     if (evflags_failed) {
00497         return k_rx_err_rtos_error;
00498     }
00499
00500     /* Clear handle */
00501     handle->initialized = false;
00502
00503     return k_rx_ok;
00504 }
```

References [rx_obstacle_detect_t::event_flags](#), [rx_obstacle_detect_t::initialized](#), [k_obstacle_detect_state_stopped](#), [rx_obstacle_detect_stop\(\)](#), [rx_obstacle_detect_t::state](#), and [rx_obstacle_detect_t::thread](#).

Here is the call graph for this function:



4.1.5.3 rx_obstacle_detect_get_state()

```
rx_err_t rx_obstacle_detect_get_state (
    const rx_obstacle_detect_t * handle,
    rx_obstacle_detect_state_t * out_state)  [nodiscard]
```

Get current detection state.

Returns the current state of the obstacle detection system.

Parameters

in	handle	Obstacle detection handle
out	out_state	Current state

Returns

k_rx_ok on success
 k_rx_err_null_ptr if handle or out_state is nullptr
 k_rx_err_invalid_state if not initialized

Get current detection state.

Returns the current operational state of the detection system by copying handle->state to *out_state. This is a cached read from handle memory; no bus or sensor transactions are issued.

Possible states:

- k_obstacle_detect_state_stopped – task suspended, not polling
- k_obstacle_detect_state_running – actively polling, no obstacle
- k_obstacle_detect_state_obstacle – obstacle confirmed by debounce

Algorithm steps:

1. Validate handle and out_state pointer are non-null
2. Verify handle is initialized
3. Copy handle->state to *out_state

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
 out_state must be a valid non-null pointer

Postcondition

*out_state contains the current state value on k_rx_ok
 handle state is unchanged

Note

Not thread-safe; state may change concurrently if detection task is running
Returns cached state – does not re-evaluate sensor readings

Example:

```
rx_obstacle_detect_state_t state;
rx_err_t err = rx_obstacle_detect_get_state(&detector, &state);
if (err == k_rx_ok && state == k_obstacle_detect_state_obstacle) {
    handle_obstacle_event();
}
```

See also

[rx_obstacle_detect_is_obstacle_detected\(\)](#) Convenience bool check
[rx_obstacle_detect_clear_obstacle\(\)](#) Clear obstacle state

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null checks, initialized state), 2 postconditions

Definition at line 768 of file [rx_obstacle_detect.c](#).

```
00770 {
00771     /* Validate inputs */
00772     if (handle == nullptr || out_state == nullptr) {
00773         return k_rx_err_null_ptr;
00774     }
00775
00776     if (!handle->initialized) {
00777         return k_rx_err_invalid_state;
00778     }
00779
00780     *out_state = handle->state;
00781
00782     return k_rx_ok;
00783 }
```

References [rx_obstacle_detect_t::initialized](#), and [rx_obstacle_detect_t::state](#).

4.1.5.4 rx_obstacle_detect_get_stats()

```
rx_err_t rx_obstacle_detect_get_stats (
    const rx_obstacle_detect_t * handle,
    uint32_t * out_total_polls,
    uint32_t * out_obstacle_events,
    uint32_t * out_false_positives) [nodiscard]
```

Get obstacle detection statistics.

Returns statistics about detection performance and events.

Parameters

in	handle	Obstacle detection handle
out	out_total_polls	Total sensor polls (can be NULL)
out	out_obstacle_events	Total obstacle events (can be NULL)
out	out_false_positives	Debounced false positives (can be NULL)

Returns

k_rx_ok on success
k_rx_err_null_ptr if handle is nullptr
k_rx_err_invalid_state if not initialized

Get obstacle detection statistics.

Returns the three running counters maintained by the detection task: total sensor poll cycles, confirmed obstacle events, and filtered false positive readings. All output parameters are optional – passing nullptr for any counter skips that output without error.

Counters are incremented by the internal detection task and are NOT reset by this function. Use [rx_obstacle_detect_reset_stats\(\)](#) to clear them.

Algorithm steps:

1. Validate handle is non-null and initialized
2. If out_total_polls != nullptr, copy handle->total_polls
3. If out_obstacle_events != nullptr, copy handle->obstacle_events
4. If out_false_positives != nullptr, copy handle->>false_positive_count

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
All non-null output pointers must be valid writable memory

Postcondition

Output values are consistent snapshots at the time of the call
Counters in the handle are unchanged

Note

Not thread-safe; counters may be incremented by detection task concurrently
All three output pointers are optional; pass nullptr for any to skip that stat

Example:

```
uint32_t polls = 0, events = 0, false_pos = 0;
rx_err_t err = rx_obstacle_detect_get_stats(&detector, &polls, &events, &>false_pos);
if (err == k_rx_ok) {
    rx_log_info("app", "Polls: %u, Obstacles: %u, FP: %u",
               (unsigned)polls, (unsigned)events, (unsigned>false_pos);
}
```

See also

[rx_obstacle_detect_reset_stats\(\)](#) Clear all counters to zero

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 881 of file [rx_obstacle_detect.c](#).

```
00885 {
00886     /* Validate inputs */
00887     if (handle == nullptr) {
00888         return k_rx_err_null_ptr;
00889     }
00890
00891     if (!handle->initialized) {
00892         return k_rx_err_invalid_state;
00893     }
00894
00895     /* Copy statistics */
00896     if (out_total_polls != nullptr) {
00897         *out_total_polls = handle->total_polls;
00898     }
00899
00900     if (out_obstacle_events != nullptr) {
00901         *out_obstacle_events = handle->obstacle_events;
00902     }
00903
00904     if (out_false_positives != nullptr) {
00905         *out_false_positives = handle->>false_positive_count;
00906     }
00907
00908     return k_rx_ok;
00909 }
```

References [rx_obstacle_detect_t::false_positive_count](#), [rx_obstacle_detect_t::initialized](#), [rx_obstacle_detect_t::obstacle_events](#) and [rx_obstacle_detect_t::total_polls](#).

4.1.5.5 rx_obstacle_detect_init()

```
rx_err_t rx_obstacle_detect_init (
    rx_obstacle_detect_t * handle,
    const rx_obstacle_detect_config_t * config) [nodiscard]
```

Initialize obstacle detection system with sensors and motors.

Initializes the obstacle detection system by creating a ThreadX task that autonomously monitors HC-SR04 sensors and triggers emergency motor stops when obstacles breach the configured safety threshold.

Initialization Sequence:

1. Validate input parameters (handle, config, sensor/motor arrays)
2. Clear handle structure (memset to zero)
3. Copy configuration parameters to handle
4. Copy sensor and motor handle pointers
5. Create ThreadX event flags for task control
6. Create ThreadX detection task (priority 8, 2048 byte stack)
7. Task starts suspended - call `rx_obstacle_detect_start()` to begin monitoring

Memory Allocation:

- Handle structure: Caller-allocated (~2.6 KB static)
- ThreadX task stack: Inside handle (2048 bytes)
- ThreadX event flags: Inside handle (~32 bytes)
- No heap allocation (`k_rx_ok`)

Parameters

out	handle	Obstacle detection handle to initialize • Must be non-NULL • Caller-allocated (declare as <code>rx_obstacle_detect_t</code>) • Must NOT be already initialized • Cleared by this function • Remains valid until <code>rx_obstacle_detect_deinit()</code>
in	config	Configuration structure • Must be non-NULL • All fields must be valid (checked by <code>internal_validate_config</code>) • Contents copied to handle (config can be stack-allocated) • Sensor/motor arrays must remain valid (only pointers copied)

Returns

Initialization result

Return values

<code>k_rx_ok</code>	System initialized successfully, ready to start
<code>k_rx_err_null_ptr</code>	handle, config, config->sensors/motors, or any sensor/motor handle in arrays is nullptr

k_rx_err_invalid_arg	sensor_count not in [1, 8]; motor_count not in [1, 4]; threshold_cm not in [2.0, 400.0]; debounce_samples not in [1, 10]; poll_interval_ms not in [10, 1000]
k_rx_err_invalid_state	handle->initialized is already true
k_rx_err_rtos_error	ThreadX event flags or task creation failed

Precondition

handle must point to valid `rx_obstacle_detect_t` storage
 config must point to valid configuration
 All sensors in config must be initialized via `rx_hcsr04_init()`
 All motors in config must be initialized via `rx_motor_init()`
 ThreadX must be running (`tx_kernel_enter()` completed)

Postcondition

On success: `handle->initialized = true`
 On success: ThreadX task created but suspended
 On success: `handle->state = k_obstacle_detect_state_stopped`
 On failure: handle unchanged (or partially cleared)
 On failure: No ThreadX resources allocated

Note

Not thread-safe - do not call concurrently on same handle
 Re-entrant for different handles
 Task starts suspended - call `rx_obstacle_detect_start()` to begin

Warning

Do not modify handle fields after initialization
 Do not deinitialize sensors/motors while detection is running
 Ensure ThreadX kernel is running before calling

Performance:

Execution time: ~100us (ThreadX task creation overhead) Memory: 2.6 KB static (handle structure with embedded stack)

Example:

```
// Declare handle (static or global recommended for large struct)
static rx_obstacle_detect_t detector;

// Setup sensor and motor arrays
rx_hcsr04_t* sensors[] = {&sensor_front, &sensor_rear};
rx_motor_handle_t* motors[] = {&motor_left, &motor_right};

// Configure detection system
rx_obstacle_detect_config_t config = {
    .sensors = sensors,
    .sensor_count = 2,
    .motors = motors,
    .motor_count = 2,
    .detection_threshold_cm = 30.0f,
    .debounce_samples = 3,
    .poll_interval_ms = 20,
    .callback = my_callback,
    .user_data = &my_context
};

// Initialize
rx_err_t ret = rx_obstacle_detect_init(&detector, &config);
if (ret != k_rx_ok) {
    printf("Init failed: %d\n", ret);
    return ret;
}

// Start monitoring
ret = rx_obstacle_detect_start(&detector);
```

See also

[rx_obstacle_detect_config_t](#) Configuration structure
[rx_obstacle_detect_start\(\)](#) Start monitoring after init
[rx_obstacle_detect_deinit\(\)](#) Cleanup and release resources
[internal_validate_config\(\)](#) Internal configuration validation

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 3: [OK] No dynamic memory - all allocation is static
- Rule 5: [OK] Minimum 6 preconditions validated
- Rule 7: [OK] All ThreadX return values checked

Definition at line 324 of file [rx_obstacle_detect.c](#).

```
00326 {
00327     /* Validate inputs */
00328     if (handle == nullptr || config == nullptr) {
00329         return k_rx_err_null_ptr;
00330     }
00331
00332     if (handle->initialized) {
00333         return k_rx_err_invalid_state;
00334     }
00335
00336     rx_err_t ret = internal_validate_config(config);
00337     if (ret != k_rx_ok) {
00338         return ret;
00339     }
00340
00341     *handle = s_zero_handle;
00342
00343     internal_copy_config(handle, config);
00344
00345     ret = internal_create_threadx_resources(handle);
```

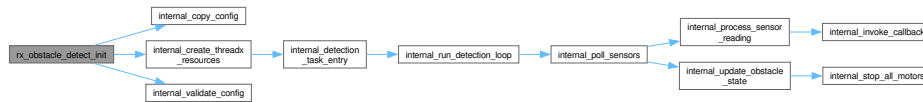
```

00346  if (ret != k_rx_ok) {
00347      return ret;
00348  }
00349
00350  handle->initialized = true;
00351
00352  return k_rx_ok;
00353 }

```

References [rx_obstacle_detect_t::initialized](#), [internal_copy_config\(\)](#), [internal_create_threadx_resources\(\)](#), [internal_validate_config\(\)](#), and [s_zero_handle](#).

Here is the call graph for this function:



4.1.5.6 rx_obstacle_detect_is_obstacle_detected()

```

bool rx_obstacle_detect_is_obstacle_detected (
    const rx_obstacle_detect_t * handle) [nodiscard]

```

Check if obstacle is currently detected.

Returns true if any sensor currently detects an obstacle within the threshold distance (after debouncing).

Parameters

in	handle	Obstacle detection handle
----	--------	---------------------------

Returns

true if obstacle detected

false if no obstacle or handle is nullptr/invalid

Check if obstacle is currently detected.

Returns true if the obstacle detection system is initialized and its current state is `k_obstacle_detect_state_obstacle`. Returns false for all other conditions including: null handle, uninitialized handle, stopped state, or running state with no obstacle.

This is a lightweight wrapper around `handle->state` that avoids allocating an output parameter, suitable for use in polling loops and conditional checks.

Algorithm steps:

1. Guard against null or uninitialized handle (returns false)
2. Return `(handle->state == k_obstacle_detect_state_obstacle)`

Precondition

None – safe to call with null handle

For meaningful results, handle should be initialized and detection started

Postcondition

handle state is unchanged

Return value reflects handle->state at the instant of the call

Note

Not thread-safe; state may change concurrently if detection task is active

Returns false for null handle rather than asserting; safe for defensive polling

Example:

```
if (rx_obstacle_detect_is_obstacle_detected(&detector)) {
    emergency_stop_all_motors();
    rx_obstacle_detect_clear_obstacle(&detector);
}
```

See also

[rx_obstacle_detect_get_state\(\)](#) Full state query with error code

[rx_obstacle_detect_clear_obstacle\(\)](#) Acknowledge and clear obstacle

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null-safe guard, initialized check), 2 postconditions

Definition at line 827 of file [rx_obstacle_detect.c](#).

```
00828 {
00829     if (handle == nullptr || !handle->initialized) {
00830         return false;
00831     }
00832
00833     return (bool)(handle->state == k_obstacle_detect_state_obstacle);
00834 }
```

References [rx_obstacle_detect_t::initialized](#), [k_obstacle_detect_state_obstacle](#), and [rx_obstacle_detect_t::state](#).

4.1.5.7 rx_obstacle_detect_reset_stats()

```
rx_err_t rx_obstacle_detect_reset_stats (
    rx_obstacle_detect_t * handle) [nodiscard]
```

Reset statistics counters.

Resets all statistics counters to zero.

Parameters

in,out	handle	Obstacle detection handle
--------	--------	---------------------------

Returns

k_rx_ok on success
 k_rx_err_null_ptr if handle is nullptr
 k_rx_err_invalid_state if not initialized

Reset statistics counters.

Clears the three running diagnostic counters maintained by the detection task: total_polls, obstacle_events, and false_positive_count. After this call all counters return to zero and begin accumulating from the next poll cycle.

Algorithm steps:

1. Validate handle is non-null and initialized
2. Set handle->total_polls to zero
3. Set handle->obstacle_events to zero
4. Set handle->>false_positive_count to zero

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
 Caller should ensure detection task is not mid-cycle to avoid partial counts

Postcondition

handle->total_polls is zero
 handle->obstacle_events is zero

Note

Not thread-safe; calling while detection task is incrementing counters may produce slightly inconsistent results; caller must synchronize if exactness required

Example:

```
// Start a fresh statistics window
rx_err_t err = rx_obstacle_detect_reset_stats(&detector);
if (err != k_rx_ok) {
    rx_log_error("app", "Failed to reset obstacle detector stats");
}
```

See also

[rx_obstacle_detect_get_stats\(\)](#) Retrieve current counter values

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 951 of file [rx_obstacle_detect.c](#).

```
00952 {
00953     /* Validate inputs */
00954     if (handle == nullptr) {
00955         return k_rx_err_null_ptr;
00956     }
00957     if (!handle->initialized) {
00958         return k_rx_err_invalid_state;
00959     }
00960 }
00961
00962 /* Reset statistics */
00963 handle->total_polls = 0;
00964 handle->obstacle_events = 0;
00965 handle->>false_positive_count = 0;
00966
00967 return k_rx_ok;
00968 }
```

References [rx_obstacle_detect_t::false_positive_count](#), [rx_obstacle_detect_t::initialized](#), [rx_obstacle_detect_t::obstacle_events](#) and [rx_obstacle_detect_t::total_polls](#).

4.1.5.8 rx_obstacle_detect_start()

```
rx_err_t rx_obstacle_detect_start (
    rx\_obstacle\_detect\_t * handle) [nodiscard]
```

Start obstacle detection monitoring.

Starts the detection task to begin polling sensors. If an obstacle is already detected, motors will be stopped immediately.

Parameters

in,out	handle	Obstacle detection handle
--------	--------	---------------------------

Returns

k_rx_ok on success
k_rx_err_null_ptr if handle is nullptr
k_rx_err_invalid_state if not initialized

Transitions the obstacle detection system to the running state by resuming the ThreadX detection thread (if it was previously suspended) and signaling the k_event_flag_start event flag to the detection task.

The detection task waits on this event flag at startup; once received it enters the polling loop and begins continuous sensor monitoring.

Algorithm steps:

1. Validate handle is non-null and initialized
2. If state is k_obstacle_detect_state_stopped, resume the ThreadX thread
3. Set k_event_flag_start event flag via tx_event_flags_set()

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
System must be in ThreadX running state (tx_kernel_enter() called)

Postcondition

Detection thread is executing and polling sensors
handle->state transitions to k_obstacle_detect_state_running

Note

Not thread-safe; caller must serialize start/stop calls
Calling start when already running is a no-op for thread resume

Example:

```
rx_err_t err = rx_obstacle_detect_start(&detector);
if (err != k_rx_ok) {
    rx_log_error("app", "Failed to start obstacle detection");
}
```

See also

[rx_obstacle_detect_stop\(\)](#) Stop detection
[rx_obstacle_detect_get_state\(\)](#) Query current detection state

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 553 of file [rx_obstacle_detect.c](#).

```

00554 {
00555     /* Validate inputs */
00556     if (handle == nullptr) {
00557         return k_rx_err_null_ptr;
00558     }
00559
00560     if (!handle->initialized) {
00561         return k_rx_err_invalid_state;
00562     }
00563
00564     /* Start thread if not already running */
00565     UINT status = TX_SUCCESS;
00566     if (handle->state == k_obstacle_detect_state_stopped) {
00567         status = tx_thread_resume(&handle->thread);
00568         if (status != TX_SUCCESS) {
00569             return k_rx_err_rtos_error;
00570         }
00571     }
00572
00573     /* Signal start event */
00574     status = tx_event_flags_set(&handle->event_flags, k_event_flag_start, TX_OR);
00575     if (status != TX_SUCCESS) {
00576         return k_rx_err_rtos_error;
00577     }
00578
00579     return k_rx_ok;
00580 }

```

References [rx_obstacle_detect_t::event_flags](#), [rx_obstacle_detect_t::initialized](#), [k_event_flag_start](#), [k_obstacle_detect_state_stopped](#), [rx_obstacle_detect_t::state](#), and [rx_obstacle_detect_t::thread](#).

4.1.5.9 rx_obstacle_detect_stop()

```

rx_err_t rx_obstacle_detect_stop (
    rx\_obstacle\_detect\_t * handle) [nodiscard]

```

Stop obstacle detection monitoring.

Stops the detection task. Motors will NOT be automatically restarted even if they were previously stopped due to obstacle detection.

Parameters

in,out	handle	Obstacle detection handle
--------	--------	---------------------------

Returns

[k_rx_ok](#) on success
[k_rx_err_null_ptr](#) if handle is nullptr
[k_rx_err_invalid_state](#) if not initialized

Requests the obstacle detection task to cease polling by setting `handle->stop_requested` and signaling the `k_event_flag_stop` event flag. The detection task checks this flag at the top of each polling iteration and transitions to the stopped state upon receipt.

This function returns immediately after signaling; it does not wait for the task to actually stop. Use [rx_obstacle_detect_get_state\(\)](#) to confirm the transition to `k_obstacle_detect_state_stopped` if synchronization is needed.

Algorithm steps:

1. Validate handle is non-null and initialized
2. Set handle->stop_requested to true
3. Set k_event_flag_stop event flag via tx_event_flags_set()

Precondition

handle must be initialized via [rx_obstacle_detect_init\(\)](#)
System must be in ThreadX running state

Postcondition

handle->stop_requested is true
k_event_flag_stop event flag is set; detection task will stop at next iteration

Note

Not thread-safe; caller must serialize start/stop calls

Warning

This function is non-blocking; the task may still be running immediately after return

Example:

```
rx_err_t err = rx_obstacle_detect_stop(&detector);
if (err != k_rx_ok) {
    rx_log_error("app", "Failed to stop obstacle detection");
}
```

See also

[rx_obstacle_detect_start\(\)](#) Resume detection
[rx_obstacle_detect_get_state\(\)](#) Poll state to confirm stop

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: 2 preconditions (null check, initialized check), 2 postconditions

Definition at line 626 of file `rx_obstacle_detect.c`.

```

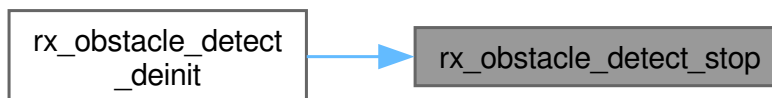
00627 {
00628     /* Validate inputs */
00629     if (handle == nullptr) {
00630         return k_rx_err_null_ptr;
00631     }
00632
00633     if (!handle->initialized) {
00634         return k_rx_err_invalid_state;
00635     }
00636
00637     /* Signal stop event */
00638     handle->stop_requested = true;
00639     UINT status = tx_event_flags_set(&handle->event_flags, k_event_flag_stop, TX_OR);
00640     if (status != TX_SUCCESS) {
00641         return k_rx_err_rtos_error;
00642     }
00643
00644     return k_rx_ok;
00645 }

```

References `rx_obstacle_detect_t::event_flags`, `rx_obstacle_detect_t::initialized`, `k_event_flag_stop`, and `rx_obstacle_detect_t::stop_requested`.

Referenced by `rx_obstacle_detect_deinit()`.

Here is the caller graph for this function:



4.2 rx_obstacle_detect.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_obstacle_detect.h
00003  * @brief Safety-Critical Obstacle Detection System with Emergency Motor Stop
00004  *
00005  * @details
00006  * Provides a ThreadX-based real-time obstacle detection system using HC-SR04
00007  * ultrasonic sensors with immediate emergency stop capability. This module
00008  * continuously monitors configured sensors and automatically stops all motors
00009  * when obstacles are detected within a configurable safety threshold.
00010  *
00011  * **System Overview:**
00012  *
00013  * This is a simplified safety system focused on collision avoidance through
00014  * emergency motor stop, not full Dynamic Window Approach (DWA) path planning.
00015  * The module runs autonomously in a ThreadX task, polling HC-SR04 sensors at
00016  * a configurable rate and triggering emergency stops when obstacles breach
00017  * the safety perimeter.
00018  *
00019  * **Architecture:**
00020  *
00021  * @dot
00022  * digraph obstacle_detect_arch {
00023  *     rankdir=TB;
00024  *     node [shape=box, style=rounded];
00025  *
00026  *     user [label="User Application"];
00027  *     api [label="rx_obstacle_detect API\n(This Module)", fillcolor=lightblue, style=filled];
00028  *     task [label="Detection ThreadX Task\n(Internal)"];
00029  *     hcsr04 [label="HC-SR04 Sensors\n(rx_hcsr04)"];
00030  *     motors [label="Motor Drivers\n(rx_motor)"];
00031  *     callback [label="Event Callback\n(Optional)", shape=ellipse];

```

```

00032 *
00033 * user -> api [label="init/start/stop"];
00034 * api -> task [label="creates/controls"];
00035 * task -> hcsr04 [label="polls @ 10-100 Hz"];
00036 * task -> motors [label="emergency stop"];
00037 * task -> callback [label="notify state change"];
00038 * callback -> user [label="obstacle events"];
00039 * }
00040 * @enddot
00041 *
00042 * **State Machine:**
00043 *
00044 * @startuml
00045 * [*] --> Stopped : init()
00046 * Stopped --> Running : start()
00047 * Running --> Obstacle : obstacle detected
00048 * Obstacle --> Running : obstacle cleared
00049 * Running --> Stopped : stop()
00050 * Obstacle --> Stopped : stop()
00051 * Stopped --> [*] : deinit()
00052 *
00053 * state Running {
00054 *   [*] --> Polling
00055 *   Polling --> Debouncing : distance < threshold
00056 *   Debouncing --> Polling : distance >= threshold
00057 *   Debouncing --> EmergencyStop : debounce_count >= samples
00058 * }
00059 * @enduml
00060 *
00061 * **Features:**
00062 *
00063 * | Feature | Description |
00064 * |-----|-----|
00065 * | **Multi-Sensor Support** | Up to 8 HC-SR04 sensors (front, sides, back) |
00066 * | **Configurable Threshold** | Stop distance 2-400cm (typical: 30cm) |
00067 * | **Debouncing** | 1-10 consecutive readings to confirm (typical: 3) |
00068 * | **Polling Rate** | 10-1000ms interval (typical: 20ms = 50Hz) |
00069 * | **ThreadX Integration** | Autonomous operation in dedicated task |
00070 * | **Event Callbacks** | Optional notification on state changes |
00071 * | **Statistics** | Track polls, events, false positives |
00072 * | **Emergency Stop** | Immediate motor shutdown on detection |
00073 *
00074 * **Detection Logic:**
00075 *
00076 * @msc
00077 * Task, Sensor1, Sensor2, Motors, Callback;
00078 *
00079 * ... [label="Normal Operation"];
00080 * Task => Sensor1 [label="measure()"];
00081 * Sensor1 => Task [label="50cm (OK)"];
00082 * Task => Sensor2 [label="measure()"];
00083 * Sensor2 => Task [label="60cm (OK)"];
00084 * Task note Task [label="wait poll_interval_ms"];
00085 *
00086 * ... [label="Obstacle Detection"];
00087 * Task => Sensor1 [label="measure()"];
00088 * Sensor1 => Task [label="25cm (< 30cm threshold)"];
00089 * Task box Task [label="debounce_counter++ (1)"];
00090 * Task note Task [label="wait poll_interval_ms"];
00091 *
00092 * Task => Sensor1 [label="measure()"];
00093 * Sensor1 => Task [label="23cm (< threshold)"];
00094 * Task box Task [label="debounce_counter++ (2)"];
00095 *
00096 * Task => Sensor1 [label="measure()"];
00097 * Sensor1 => Task [label="22cm (< threshold)"];
00098 * Task box Task [label="debounce_counter++ (3 = threshold!)"];
00099 * Task => Motors [label="EMERGENCY STOP"];
00100 * Motors box Motors [label="All motors stopped"];
00101 * Task => Callback [label="obstacle_detected=true"];
00102 * Task box Task [label="state = OBSTACLE"];
00103 * @endmsc
00104 *
00105 * **Performance Characteristics:**
00106 *
00107 * | Parameter | Value | Notes |
00108 * |-----|-----|-----|
00109 * | **Detection Latency** | poll_interval_ms x debounce_samples | e.g., 20ms x 3 = 60ms worst-case |
00110 * | **CPU Usage** | ~5% @ 50Hz, 4 sensors | RX72N @ 240 MHz |
00111 * | **Memory (Static)** | ~2.5 KB | Stack (2048) + handle struct (512) |
00112 * | **Memory (Dynamic)** | 0 | Zero heap allocation |
00113 * | **Thread Priority** | 8 (configurable) | Higher = more responsive |
00114 *
00115 * **Safety Considerations:**
00116 *
00117 * @warning This is a last-resort safety system, NOT primary collision avoidance
00118 * @warning Detection latency depends on polling rate and debounce settings

```

```

00119 * @warning Sensors have blind spots - physical robot design must account for this
00120 * @warning At maximum polling rate (100Hz, 1 debounce), minimum latency is 10ms
00121 * @warning At typical settings (50Hz, 3 debounce), worst-case latency is 60ms
00122 *
00123 * @attention Robot traveling at 1 m/s moves 6cm during worst-case 60ms detection
00124 * @attention Always leave safety margin: threshold > max_travel_during_latency
00125 * @attention For 1 m/s max speed, use threshold >= 10cm (60ms x 1m/s + margin)
00126 *
00127 * @par Hardware Requirements:
00128 *
00129 * | Resource | Requirement | Notes |
00130 * |-----|-----|-----|
00131 * | **CPU** | RX72N @ 240 MHz | ~12k CPU cycles per poll (4 sensors) |
00132 * | **RAM** | 2.5 KB static | Task stack + handle structure |
00133 * | **ThreadX** | Required | Task creation and event flags |
00134 * | **HC-SR04 Sensors** | 1-8 sensors | Each requires 2 GPIO pins |
00135 * | **Motor Controllers** | 1-4 motors | Must support emergency stop |
00136 *
00137 * @par Module Dependencies:
00138 * - `rx_hcsr04`: HC-SR04 ultrasonic sensor driver
00139 * - `rx_motor`: Motor control with emergency stop capability
00140 * - `rx_err`: Error code definitions
00141 * - `tx_api`: ThreadX RTOS for task management
00142 * - `rx_check`: Assertion and validation macros
00143 * - `rx_threadx_config`: ThreadX configuration constants
00144 *
00145 * @par NASA Power of 10 Compliance:
00146 *
00147 * | Rule | Status | Implementation |
00148 * |-----|-----|-----|
00149 * | 1. Simple control flow | [OK] | No goto, setjmp, recursion |
00150 * | 2. Fixed loop bounds | [OK] | All loops use enum constants |
00151 * | 3. No dynamic memory | [OK] | Zero malloc/free - static allocation only |
00152 * | 4. Functions <=60 lines | [OK] | Longest: `internal_poll_sensors` (54 lines) |
00153 * | 5. Min 2 assertions/func | [OK] | All public APIs validate inputs |
00154 * | 6. Smallest scope | [OK] | Variables declared at first use |
00155 * | 7. Check return values | [OK] | All HC-SR04 and motor calls checked |
00156 * | 8. Limit preprocessor | [OK] | Only C23 typed enums, no macros |
00157 * | 9. Restrict pointers | [OK] | Max one level of dereferencing |
00158 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00159 *
00160 * @par SOLID Principles:
00161 *
00162 * **Single Responsibility (S):**
00163 * - Module has ONE job: Monitor sensors and stop motors on obstacle detection
00164 * - Does NOT handle path planning, navigation, or motor control logic
00165 * - Does NOT configure sensors or motors - only uses them
00166 *
00167 * **Open/Closed (O):**
00168 * - Extensible via configuration (threshold, debounce, polling rate)
00169 * - Open for extension via callback mechanism
00170 * - Closed for modification - core logic is stable
00171 *
00172 * **Dependency Inversion (D):**
00173 * - Depends on `rx_hcsr04` and `rx_motor` abstractions, not implementations
00174 * - Accepts sensor/motor handles via dependency injection (config)
00175 * - Callback mechanism allows user code to react without modifying module
00176 *
00177 * @see rx_hcsr04.h HC-SR04 ultrasonic sensor driver
00178 * @see rx_motor.h Motor control with emergency stop
00179 * @see rx_obstacle_detect.c Implementation file
00180 *
00181 * @author Locked, Inc.
00182 * @date 2026-01-27
00183 * @copyright Copyright (c) 2026 Locked Inc.
00184 * SPDX-License-Identifier: MIT
00185 * @version 1.0.0
00186 *
00187 * @since Version 1.0.0
00188 */
00189
00190 #pragma once
00191
00192 #include <stdbool.h>
00193 #include <stdint.h>
00194
00195 #include "rx_err.h"
00196 #include "rx_hcsr04.h"
00197 #include "rx_motor.h"
00198 #include "tx_api.h"
00199
00200 #ifdef __cplusplus
00201 extern "C" {
00202 #endif
00203
00204 /*
=====

```

```

00205 * Constants
00206 *
=====
00207 */
00208
00209 /**
00210 * @enum rx_obstacle_detect_constants_t
00211 * @brief Obstacle detection system configuration constants
00212 *
00213 * @details
00214 * Defines limits and defaults for the obstacle detection system including
00215 * maximum sensor/motor counts, ThreadX task configuration, and resource limits.
00216 *
00217 * @par Value Rationale:
00218 *
00219 * | Constant | Value | Rationale |
00220 * |-----|-----|-----|
00221 * | max_sensors | 8 | Typical robot: 4 front, 2 sides, 2 rear |
00222 * | max_motors | 4 | STAR robot has 4 drive motors |
00223 * | task_stack_size | 2048 bytes | Sufficient for nested calls + local buffers |
00224 * | task_priority | 8 | Medium priority (0=highest, 31=lowest) |
00225 *
00226 * **Task Priority Considerations:**
00227 * - Priority 8 balances responsiveness with system load
00228 * - Higher priority (lower number) reduces detection latency
00229 * - Lower priority (higher number) reduces CPU contention
00230 * - Typical ThreadX priorities: 0-7 critical, 8-15 high, 16-23 normal, 24-31 low
00231 *
00232 * @see rx_obstacle_detect_config_t Uses these limits for validation
00233 * @see rx_obstacle_detect_t Allocates arrays sized by these constants
00234 *
00235 * @since Version 1.0.0
00236 */
00237 typedef enum : uint16_t {
00238 /**< @brief Maximum number of HC-SR04 sensors supported simultaneously
00239 *
00240 * @details
00241 * Maximum sensors that can be monitored in a single detection instance.
00242 * Each sensor requires 2 GPIO pins (TRIG, ECHO) and ~64 bytes RAM.
00243 *
00244 * @par Value: 8 sensors
00245 * @par Memory Impact: 8 x 64 bytes = 512 bytes (handle struct arrays)
00246 * @par Typical Usage: 4-6 sensors for 360deg coverage
00247 */
00248 k_obstacle_detect_max_sensors = 8,
00249
00250 /**< @brief Maximum number of motors controlled by emergency stop
00251 *
00252 * @details
00253 * Maximum motors that will be stopped when obstacle detected. Each motor
00254 * is an `rx_motor_handle_t` pointer in configuration.
00255 *
00256 * @par Value: 4 motors
00257 * @par Memory Impact: 4 x sizeof(rx_motor_handle_t*) = 16 bytes
00258 * @par STAR Robot: Uses all 4 motors (front-left, front-right, rear-left, rear-right)
00259 */
00260 k_obstacle_detect_max_motors = 4,
00261
00262 /**< @brief Detection task stack size in bytes
00263 *
00264 * @details
00265 * Stack memory allocated for the ThreadX detection task. Must be sufficient
00266 * for deepest call chain plus local variables and ThreadX overhead.
00267 *
00268 * @par Value: 2048 bytes (2 KB)
00269 *
00270 * @par Stack Usage Analysis:
00271 * - ThreadX overhead: ~256 bytes
00272 * - Task entry function: ~64 bytes
00273 * - internal_poll_sensors(): ~128 bytes
00274 * - rx_hcsr04_measure_blocking(): ~256 bytes (deepest call)
00275 * - rx_motor_stop(): ~128 bytes
00276 * - Margin: ~1216 bytes (safe margin for future changes)
00277 *
00278 * @par Rationale:
00279 * 2048 bytes provides ~4x safety margin over measured usage (~512 bytes).
00280 * ThreadX stack overflow detection will catch underruns during testing.
00281 *
00282 * @warning Insufficient stack causes undefined behavior (crashes, corruption)
00283 */
00284 k_obstacle_detect_task_stack_size = 2048,
00285
00286 /**< @brief Detection task ThreadX priority level
00287 *
00288 * @details
00289 * ThreadX task priority for obstacle detection. Lower number = higher priority.
00290 * Priority 8 provides good responsiveness without starving other tasks.

```

```

00291 *
00292 * @par Value: 8 (medium-high priority)
00293 *
00294 * @par Priority Levels in STAR System:
00295 * - 0-3: Critical interrupt handlers
00296 * - 4-7: Safety-critical tasks (e.g., motor control PID)
00297 * - **8-11: High-priority tasks (obstacle detection)**
00298 * - 12-15: Normal application tasks
00299 * - 16-31: Background/low-priority tasks
00300 *
00301 * @par Rationale:
00302 * Priority 8 ensures:
00303 * - Obstacle detection preempts normal tasks
00304 * - Does not starve motor control (priority 5)
00305 * - Typical 20ms poll interval leaves CPU time for other tasks
00306 *
00307 * @par Performance Impact:
00308 * - @ 50Hz polling (20ms), task runs ~1ms/poll = 5% CPU @ 240 MHz
00309 * - Leaves 95% CPU for other tasks
00310 *
00311 * @see tx_thread_create() ThreadX task creation uses this priority
00312 */
00313 k_obstacle_detect_task_priority = 8,
00314 } rx_obstacle_detect_constants_t;
00315
00316 /**
00317 * @enum rx_obstacle_detect_state_t
00318 * @brief Obstacle detection system operational states
00319 *
00320 * @details
00321 * Defines the three possible states of the obstacle detection system.
00322 * State transitions are controlled by user commands (start/stop) and
00323 * sensor measurements (obstacle detected/cleared).
00324 *
00325 * **State Transition Diagram:**
00326 *
00327 * @startuml
00328 * [*] --> STOPPED : rx_obstacle_detect_init()
00329 *
00330 * STOPPED --> RUNNING : rx_obstacle_detect_start()
00331 *
00332 * RUNNING --> OBSTACLE : any sensor < threshold\n(after debouncing)
00333 * RUNNING --> STOPPED : rx_obstacle_detect_stop()
00334 *
00335 * OBSTACLE --> RUNNING : all sensors >= threshold\n(obstacle cleared)
00336 * OBSTACLE --> STOPPED : rx_obstacle_detect_stop()
00337 *
00338 * STOPPED --> [*] : rx_obstacle_detect_deinit()
00339 *
00340 * note right of STOPPED
00341 *   Task suspended
00342 *   No polling active
00343 *   Motors controllable
00344 * end note
00345 *
00346 * note right of RUNNING
00347 *   Task active
00348 *   Polling sensors @ poll_interval_ms
00349 *   Motors controllable
00350 * end note
00351 *
00352 * note right of OBSTACLE
00353 *   Task active
00354 *   Polling continues
00355 *   **Motors STOPPED (emergency)**
00356 * end note
00357 * @enduml
00358 *
00359 * **State Transition Table:**
00360 *
00361 * | Current State | Event | Next State | Action | Guard |
00362 * |-----|-----|-----|-----|-----|
00363 * | STOPPED | start() | RUNNING | Resume ThreadX task | initialized == true |
00364 * | RUNNING | distance < threshold | OBSTACLE | Emergency stop motors | debounce_counter >= samples |
00365 * | RUNNING | stop() | STOPPED | Suspend task | none |
00366 * | OBSTACLE | distance >= threshold | RUNNING | (motors remain stopped) | all sensors clear |
00367 * | OBSTACLE | stop() | STOPPED | Suspend task | none |
00368 * | STOPPED | clear_obstacle() | STOPPED | Clear debounce state | none (manual override) |
00369 *
00370 * @par State Descriptions:
00371 * - **STOPPED**: System initialized but not monitoring. ThreadX task suspended. Motors controllable.
00372 * - **RUNNING**: Active monitoring, no obstacles. Task polling sensors at configured rate. Motors controllable.
00373 * - **OBSTACLE**: Obstacle within threshold. Motors emergency-stopped. Monitoring continues for clearance.
00374 *
00375 * @see rx_obstacle_detect_get_state() Query current state
00376 * @see rx_obstacle_detect_start() Transition STOPPED -> RUNNING
00377 * @see rx_obstacle_detect_stop() Transition to STOPPED

```

```

00378 * @see internal_poll_sensors() Implements RUNNING -> OBSTACLE transition
00379 *
00380 * @since Version 1.0.0
00381 */
00382 typedef enum : uint8_t {
00383     /**< @brief Detection system stopped - not monitoring sensors
00384     *
00385     * @details
00386     * Initial state after `rx_obstacle_detect_init()`. ThreadX detection task
00387     * is created but suspended. No sensor polling occurs. Motors are controllable
00388     * by application.
00389     *
00390     * @par Entry Actions:
00391     * - Suspend ThreadX detection task
00392     * - Clear event flags
00393     *
00394     * @par Do Actions:
00395     * - None (task suspended)
00396     *
00397     * @par Exit Actions:
00398     * - None
00399     *
00400     * @par Valid Transitions:
00401     * - STOPPED -> RUNNING: via `rx_obstacle_detect_start()`
00402     *
00403     * @par Value: 0 (default/initial state)
00404     */
00405     k_obstacle_detect_state_stopped = 0,
00406
00407     /**< @brief Detection active - monitoring sensors, no obstacle detected
00408     *
00409     * @details
00410     * Normal operational state. ThreadX task actively polling all configured
00411     * sensors at `poll_interval_ms` rate. No obstacles within threshold distance.
00412     * Motors controllable by application.
00413     *
00414     * @par Entry Actions:
00415     * - Resume ThreadX detection task
00416     * - Set event flags (start flag)
00417     * - Reset debounce counters
00418     *
00419     * @par Do Actions:
00420     * - Poll each sensor via `rx_hcsr04_measure_blocking()`
00421     * - Check distance against threshold
00422     * - Increment debounce counter if distance < threshold
00423     * - Clear debounce counter if distance >= threshold
00424     * - Sleep for poll_interval_ms between polls
00425     *
00426     * @par Exit Actions:
00427     * - Suspend task (if stop requested)
00428     * - OR emergency stop motors (if obstacle confirmed)
00429     *
00430     * @par Valid Transitions:
00431     * - RUNNING -> OBSTACLE: When debounce_counter >= debounce_samples
00432     * - RUNNING -> STOPPED: via `rx_obstacle_detect_stop()`
00433     *
00434     * @par Value: 1
00435     */
00436     k_obstacle_detect_state_running = 1,
00437
00438     /**< @brief Obstacle detected - motors emergency stopped
00439     *
00440     * @details
00441     * Critical state indicating obstacle within safety threshold. All configured
00442     * motors have been emergency-stopped. Monitoring continues to detect when
00443     * obstacle clears. Motors cannot be controlled until obstacle clears or
00444     * user calls `rx_obstacle_detect_clear_obstacle()`.
00445     *
00446     * @par Entry Actions:
00447     * - Call `internal_stop_all_motors()` (emergency stop)
00448     * - Invoke user callback with obstacle_detected=true
00449     * - Increment obstacle_events counter
00450     *
00451     * @par Do Actions:
00452     * - Continue polling sensors
00453     * - Check if ALL sensors clear (distance >= threshold)
00454     * - Monitor for user stop request
00455     *
00456     * @par Exit Actions:
00457     * - Invoke user callback with obstacle_detected=false (if clearing)
00458     * - OR suspend task (if stop requested)
00459     *
00460     * @par Valid Transitions:
00461     * - OBSTACLE -> RUNNING: When all sensors >= threshold (automatic)
00462     * - OBSTACLE -> STOPPED: via `rx_obstacle_detect_stop()`
00463     *
00464     * @par Value: 2

```

```

00465 *
00466 * @warning Motors remain stopped until obstacle clears or manual override
00467 * @attention Application must NOT attempt to control motors in this state
00468 */
00469 k_obstacle_detect_state_obstacle = 2,
00470 } rx_obstacle_detect_state_t;
00471
00472 /*
=====
00473 * Type Definitions
00474 *
=====
00475 */
00476
00477 /**
00478 * @typedef rx_obstacle_detect_callback_t
00479 * @brief User callback function for obstacle detection state change events
00480 *
00481 * @details
00482 * Optional callback invoked by the detection task when obstacle state changes:
00483 * - Obstacle detected (distance < threshold after debouncing)
00484 * - Obstacle cleared (distance >= threshold)
00485 *
00486 * The callback is invoked from the ThreadX detection task context. Keep
00487 * processing brief to avoid delaying sensor polling. Do NOT perform blocking
00488 * operations or lengthy computations in the callback.
00489 *
00490 * **Callback Execution Context:**
00491 * - Called from: ThreadX obstacle detection task
00492 * - Task priority: 8 (k_obstacle_detect_task_priority)
00493 * - Stack: Detection task stack (2048 bytes)
00494 * - Interrupt level: Task level (not ISR)
00495 *
00496 * @par Typical Use Cases:
00497 * - Log obstacle events to console or storage
00498 * - Update UI/telemetry with obstacle status
00499 * - Trigger additional safety responses
00500 * - Notify higher-level navigation system
00501 * - Record statistics or diagnostics
00502 *
00503 * @param[in] obstacle_detected Obstacle state change direction
00504 * - true: Obstacle detected (entering OBSTACLE state)
00505 * - false: Obstacle cleared (returning to RUNNING state)
00506 * - Type: bool
00507 *
00508 * @param[in] sensor_idx Index of sensor that triggered this event
00509 * - Range: [0, sensor_count-1]
00510 * - Type: uint8_t
00511 * - For detected events: Sensor that first exceeded debounce threshold
00512 * - For cleared events: Last sensor that cleared
00513 * - Maps to sensors[] array in rx_obstacle_detect_config_t
00514 *
00515 * @param[in] distance_cm Measured distance in centimeters at time of event
00516 * - Range: [2.0, 400.0] cm (HC-SR04 valid range)
00517 * - Type: float
00518 * - Unit: centimeters
00519 * - For detected events: Final distance that triggered (< threshold)
00520 * - For cleared events: Distance when sensor cleared (>= threshold)
00521 * - May be slightly different from threshold due to polling intervals
00522 *
00523 * @param[in] user_data User-provided context pointer from configuration
00524 * - Type: void* (caller-defined)
00525 * - Passed through from rx_obstacle_detect_config_t.user_data
00526 * - Can be NULL if no context needed
00527 * - Typical usage: Pointer to application state structure
00528 *
00529 * @pre Callback is called from ThreadX task context (not ISR)
00530 * @pre sensor_idx < sensor_count (validated before callback)
00531 * @pre distance_cm within HC-SR04 valid range
00532 *
00533 * @post Callback should return promptly (< 1ms recommended)
00534 * @post Must not call blocking functions (sleep, pend, etc.)
00535 * @post Must not call obstacle detection APIs (would deadlock)
00536 *
00537 * @note Thread Safety: Callback may be interrupted by higher-priority tasks
00538 * @note Re-entrancy: Will NOT be called recursively (single detection task)
00539 * @note Null Handling: If callback is nullptr in config, no callback invoked
00540 *
00541 * @warning DO NOT block or delay in callback - delays sensor polling
00542 * @warning DO NOT call rx_obstacle_detect_* APIs from callback - causes deadlock
00543 * @warning DO NOT perform heavy computation - run on detection task stack
00544 *
00545 * @par Example - Logging Callback:
00546 * @code{.c}
00547 * void obstacle_event_callback(bool obstacle_detected, uint8_t sensor_idx,
00548 *                             float distance_cm, void* user_data) {
00549 *     if (obstacle_detected) {

```



```

00550 *      printf("OBSTACLE! Sensor %u: %.1f cm\n", sensor_idx, distance_cm);
00551 *    } else {
00552 *      printf("Cleared. Sensor %u: %.1f cm\n", sensor_idx, distance_cm);
00553 *    }
00554 *  }
00555 * @endcode
00556 *
00557 * @par Example - Telemetry Update:
00558 * @code{.c}
00559 * typedef struct {
00560 *   telemetry_t* telem;
00561 *   uint32_t     event_count;
00562 * } app_context_t;
00563 *
00564 * void obstacle_telemetry_callback(bool obstacle_detected, uint8_t sensor_idx,
00565 *                                  float distance_cm, void* user_data) {
00566 *   app_context_t* ctx = (app_context_t*)user_data;
00567 *   ctx->event_count++;
00568 *   ctx->telem->obstacle_active = obstacle_detected;
00569 *   ctx->telem->min_distance_cm = distance_cm;
00570 * }
00571 * @endcode
00572 *
00573 * @see rx_obstacle_detect_config_t Configuration includes callback and user_data
00574 * @see rx_obstacle_detect_init() Registers callback during initialization
00575 * @see internal_poll_sensors() Invokes callback on state changes
00576 *
00577 * @since Version 1.0.0
00578 */
00579 typedef void (*rx_obstacle_detect_callback_t)(bool obstacle_detected,
00580                                                uint8_t sensor_idx,
00581                                                float distance_cm,
00582                                                void* user_data);
00583
00584 /**
00585 * @struct rx_obstacle_detect_config_t
00586 * @brief Obstacle detection system configuration parameters
00587 *
00588 * @details
00589 * Configuration structure passed to `rx_obstacle_detect_init()` to set up
00590 * the detection system. Defines which sensors and motors to use, detection
00591 * thresholds, debouncing behavior, and optional event callbacks.
00592 *
00593 * **Configuration Guidelines:**
00594 *
00595 * | Parameter | Typical Range | Recommended | Notes |
00596 * |-----|-----|-----|-----|
00597 * | sensor_count | 1-8 | 4 | Front coverage minimum |
00598 * | motor_count | 1-4 | 4 | All drive motors |
00599 * | threshold_cm | 10-100 | 30 | Depends on max speed |
00600 * | debounce_samples | 1-5 | 3 | Balance latency vs false positives |
00601 * | poll_interval_ms | 10-100 | 20 | 50Hz good balance |
00602 *
00603 * **Threshold Selection Formula:**
00604 *
00605 * Choose threshold based on maximum robot velocity and acceptable stopping distance:
00606 *
00607 * @f[
00608 * \text{threshold}_{cm} \geq v_{max} \times t_{latency} + d_{stop} + \text{margin}
00609 * @f]
00610 *
00611 * where:
00612 * - @f$ v_{max} @f$ = maximum robot velocity (cm/s)
00613 * - @f$ t_{latency} @f$ = detection latency (s) = poll_interval x debounce_samples
00614 * - @f$ d_{stop} @f$ = braking distance (cm)
00615 * - margin = safety factor (typically 5-10cm)
00616 *
00617 * **Example Calculation:**
00618 * - Max velocity: 100 cm/s (1 m/s)
00619 * - Poll interval: 20ms, Debounce: 3 -> Latency = 60ms = 0.06s
00620 * - Travel during latency: 100 x 0.06 = 6cm
00621 * - Braking distance: ~10cm (depends on motor/PID)
00622 * - Margin: 10cm
00623 * - **Threshold: 6 + 10 + 10 = 26cm -> use 30cm**
00624 *
00625 * @par Field Constraints:
00626 *
00627 * | Field | Constraint | Validation |
00628 * |-----|-----|-----|
00629 * | sensors | Non-nullptr | Checked in init |
00630 * | sensor_count | [1, 8] | Checked in init |
00631 * | motors | Non-nullptr | Checked in init |
00632 * | motor_count | [1, 4] | Checked in init |
00633 * | threshold_cm | [2.0, 400.0] | HC-SR04 range |
00634 * | debounce_samples | [1, 10] | Checked in init |
00635 * | poll_interval_ms | [10, 1000] | Checked in init |
00636 * | callback | Can be NULL | Optional |

```

```

00637 * | user_data | Can be NULL | Optional |
00638 *
00639 * @par Usage Example:
00640 * @code{.c}
00641 * // Setup sensors
00642 * rx_hcsr04_t sensor_front, sensor_left, sensor_right, sensor_rear;
00643 * rx_hcsr04_t* sensors[] = {&sensor_front, &sensor_left, &sensor_right, &sensor_rear};
00644 *
00645 * // Setup motors
00646 * rx_motor_handle_t motor_fl, motor_fr, motor_rl, motor_rr;
00647 * rx_motor_handle_t* motors[] = {&motor_fl, &motor_fr, &motor_rl, &motor_rr};
00648 *
00649 * // Configure obstacle detection
00650 * rx_obstacle_detect_config_t config = {
00651 *     .sensors = sensors,
00652 *     .sensor_count = 4,
00653 *     .motors = motors,
00654 *     .motor_count = 4,
00655 *     .detection_threshold_cm = 30.0f, // 30cm safety perimeter
00656 *     .debounce_samples = 3,           // 3 consecutive readings
00657 *     .poll_interval_ms = 20,          // 50Hz polling
00658 *     .callback = obstacle_event_handler,
00659 *     .user_data = &app_context
00660 * };
00661 *
00662 * rx_obstacle_detect_t detector;
00663 * rx_err_t ret = rx_obstacle_detect_init(&detector, &config);
00664 * @endcode
00665 *
00666 * @see rx_obstacle_detect_init() Uses this configuration
00667 * @see internal_validate_config() Validates field constraints
00668 *
00669 * @since Version 1.0.0
00670 */
00671 typedef struct {
00672     /**< @brief Array of pointers to initialized HC-SR04 sensor handles
00673     *
00674     * @details
00675     * Array of sensor handles to monitor for obstacles. Each sensor must be
00676     * pre-initialized via `rx_hcsr04_init()` before passing to obstacle detection.
00677     *
00678     * @par Requirements:
00679     * - Must be non-NULL
00680     * - Must contain sensor_count valid pointers
00681     * - Each element must point to initialized rx_hcsr04_t
00682     * - Array must remain valid for lifetime of detection system
00683     *
00684     * @par Typical Configuration:
00685     * - 4 sensors: Front, rear, left, right (90deg coverage)
00686     * - 6 sensors: Frontx2, sidesx2, rearsx2 (180deg coverage)
00687     * - 8 sensors: Full 360deg coverage
00688     *
00689     * @warning Sensors must remain initialized while detection is running
00690     */
00691     rx_hcsr04_t** sensors;
00692
00693     /**< @brief Number of sensors in sensors array
00694     *
00695     * @details
00696     * Count of valid sensor pointers in the sensors array.
00697     *
00698     * @par Valid Range: [1, k_obstacle_detect_max_sensors] (1-8)
00699     * @par Typical Value: 4 (front/rear/left/right)
00700     *
00701     * @warning Must match actual sensors array size
00702     */
00703     uint8_t sensor_count;
00704
00705     /**< @brief Array of pointers to initialized motor control handles
00706     *
00707     * @details
00708     * Array of motor handles to stop when obstacle detected. Each motor must
00709     * be pre-initialized via `rx_motor_init()` before passing to obstacle detection.
00710     *
00711     * @par Requirements:
00712     * - Must be non-NULL
00713     * - Must contain motor_count valid pointers
00714     * - Each element must point to initialized rx_motor_handle_t
00715     * - Array must remain valid for lifetime of detection system
00716     *
00717     * @par Typical Configuration:
00718     * - 2 motors: Differential drive (left, right)
00719     * - 4 motors: Four-wheel drive (FL, FR, RL, RR)
00720     *
00721     * @warning Motors must remain initialized while detection is running
00722     */
00723     rx_motor_handle_t** motors;

```

```

00724
00725 /**< @brief Number of motors in motors array
00726 *
00727 * @details
00728 * Count of valid motor pointers in the motors array.
00729 *
00730 * @par Valid Range: [1, k_obstacle_detect_max_motors] (1-4)
00731 * @par Typical Value: 4 (four-wheel drive)
00732 *
00733 * @warning Must match actual motors array size
00734 */
00735 uint8_t motor_count;
00736
00737 /**< @brief Obstacle detection threshold distance in centimeters
00738 *
00739 * @details
00740 * Minimum safe distance from obstacles. If any sensor measures distance
00741 * below this threshold (after debouncing), emergency motor stop is triggered.
00742 *
00743 * @par Valid Range: [2.0, 400.0] cm (HC-SR04 specification)
00744 * @par Typical Values:
00745 * - 10cm: Very close operation (indoor, slow speeds)
00746 * - 30cm: General purpose (outdoor, moderate speeds)
00747 * - 50cm: High-speed operation (fast robots)
00748 *
00749 * @par Unit: Centimeters (cm)
00750 *
00751 * @par Selection Guidelines:
00752 * threshold >= (max_speed_cm/s x latency_s) + braking_distance_cm + 10cm margin
00753 *
00754 * @warning Too small: Insufficient stopping time -> collisions
00755 * @warning Too large: Excessive false stops, limited mobility
00756 */
00757 float detection_threshold_cm;
00758
00759 /**< @brief Number of consecutive sensor readings required to confirm obstacle
00760 *
00761 * @details
00762 * Debounce count to filter out transient sensor noise or spurious readings.
00763 * Obstacle is only confirmed after this many consecutive readings below threshold.
00764 *
00765 * @par Valid Range: [1, 10]
00766 * @par Typical Values:
00767 * - 1: No debouncing (fastest response, more false positives)
00768 * - 3: Good balance (recommended)
00769 * - 5: Heavy debouncing (slower response, fewer false positives)
00770 *
00771 * @par Impact on Latency:
00772 * detection_latency = poll_interval_ms x debounce_samples
00773 * - 20ms x 3 = 60ms latency
00774 * - 10ms x 3 = 30ms latency
00775 *
00776 * @warning Higher values increase detection latency
00777 */
00778 uint32_t debounce_samples;
00779
00780 /**< @brief Sensor polling interval in milliseconds
00781 *
00782 * @details
00783 * Time delay between sensor measurement cycles. Determines polling rate
00784 * and CPU usage.
00785 *
00786 * @par Valid Range: [10, 1000] ms
00787 * @par Typical Values:
00788 * - 10ms: 100Hz (very responsive, higher CPU usage)
00789 * - 20ms: 50Hz (good balance, recommended)
00790 * - 50ms: 20Hz (slower response, lower CPU usage)
00791 *
00792 * @par Unit: Milliseconds (ms)
00793 *
00794 * @par CPU Impact:
00795 * - 100Hz (10ms): ~10% CPU @ 4 sensors
00796 * - 50Hz (20ms): ~5% CPU @ 4 sensors
00797 * - 20Hz (50ms): ~2% CPU @ 4 sensors
00798 *
00799 * @par Latency Impact:
00800 * Worst-case detection latency = poll_interval_ms x debounce_samples
00801 *
00802 * @warning Lower intervals increase CPU load
00803 * @warning Higher intervals increase detection latency
00804 */
00805 uint32_t poll_interval_ms;
00806
00807 /**< @brief Optional callback for obstacle state change events
00808 *
00809 * @details
00810 * Function pointer invoked when obstacles are detected or cleared.

```

```

00811  * Set to NULL if no callback needed.
00812  *
00813  * @par Valid Values: Function pointer or nullptr
00814  * @par Typical Usage: Logging, telemetry updates, UI notifications
00815  *
00816  * @see rx_obstacle_detect_callback_t Callback signature
00817  *
00818  * @warning Callback invoked from detection task - keep processing brief
00819  */
00820 rx_obstacle_detect_callback_t callback;
00821
00822 /**< @brief User-defined context pointer passed to callback
00823  *
00824  * @details
00825  * Arbitrary user data passed through to callback function. Typically
00826  * points to application state structure. Set to NULL if not needed.
00827  *
00828  * @par Valid Values: Any pointer or nullptr
00829  * @par Lifetime: Must remain valid while detection is running
00830  *
00831  * @par Typical Usage:
00832  * - Pointer to application context struct
00833  * - Pointer to telemetry buffer
00834  * - Pointer to logging state
00835  *
00836  * @warning If non-NULL, must remain valid until deinit
00837  */
00838 void* user_data;
00839 } rx_obstacle_detect_config_t;
00840
00841 /**
00842  * @brief Obstacle detection handle
00843  *
00844  * Maintains detection state and statistics. Caller allocates and passes to
00845  * init. Do not modify fields directly after initialization.
00846  */
00847 typedef struct {
00848     /* Configuration (copied from config) */
00849     rx_hcsr04_t*          sensors[k_obstacle_detect_max_sensors]; /**< Sensor handles */
00850     uint8_t              sensor_count; /**< Number of sensors */
00851     rx_motor_handle_t*    motors[k_obstacle_detect_max_motors]; /**< Motor handles */
00852     uint8_t              motor_count; /**< Number of motors */
00853     float                detection_threshold_cm; /**< Stop distance threshold */
00854     uint32_t             debounce_samples; /**< Debounce sample count */
00855     uint32_t             poll_interval_ms; /**< Polling interval */
00856     rx_obstacle_detect_callback_t callback; /**< Event callback */
00857     void*                user_data; /**< User context */
00858
00859     /* ThreadX resources */
00860     TX_THREAD            thread; /**< Detection task thread */
00861     uint8_t              thread_stack[k_obstacle_detect_task_stack_size]; /**< Task stack */
00862     TX_EVENT_FLAGS_GROUP event_flags; /**< Control event flags */
00863
00864     /* State */
00865     bool                 initialized; /**< True if initialized */
00866     rx_obstacle_detect_state_t state; /**< Current detection state */
00867     bool                 stop_requested; /**< True if stop requested */
00868
00869     /* Debouncing state (per-sensor) */
00870     uint32_t             debounce_counter[k_obstacle_detect_max_sensors]; /**< Consecutive detections */
00871     bool                 obstacle_active[k_obstacle_detect_max_sensors]; /**< Per-sensor obstacle state */
00872
00873     /* Statistics */
00874     uint32_t             total_polls; /**< Total sensor polls performed */
00875     uint32_t             obstacle_events; /**< Total obstacle detection events */
00876     uint32_t             false_positive_count; /**< Debounced false positives */
00877 } rx_obstacle_detect_t;
00878
00879 /*
=====
00880  * Public API - Initialization
00881  *
=====
00882  */
00883
00884 /**
00885  * @brief Initialize obstacle detection system with sensors and motors
00886  *
00887  * @details
00888  * Initializes the obstacle detection system by creating a ThreadX task that
00889  * autonomously monitors HC-SR04 sensors and triggers emergency motor stops
00890  * when obstacles breach the configured safety threshold.
00891  *
00892  * **Initialization Sequence:**
00893  * 1. Validate input parameters (handle, config, sensor/motor arrays)
00894  * 2. Clear handle structure (memset to zero)
00895  * 3. Copy configuration parameters to handle

```

```

00896 * 4. Copy sensor and motor handle pointers
00897 * 5. Create ThreadX event flags for task control
00898 * 6. Create ThreadX detection task (priority 8, 2048 byte stack)
00899 * 7. Task starts suspended - call rx_obstacle_detect_start() to begin monitoring
00900 *
00901 * **Memory Allocation:**
00902 * - Handle structure: Caller-allocated (~2.6 KB static)
00903 * - ThreadX task stack: Inside handle (2048 bytes)
00904 * - ThreadX event flags: Inside handle (~32 bytes)
00905 * - No heap allocation (k_rx_ok)
00906 *
00907 * @param[out] handle Obstacle detection handle to initialize
00908 * - Must be non-NULL
00909 * - Caller-allocated (declare as rx_obstacle_detect_t)
00910 * - Must NOT be already initialized
00911 * - Cleared by this function
00912 * - Remains valid until rx_obstacle_detect_deinit()
00913 *
00914 * @param[in] config Configuration structure
00915 * - Must be non-NULL
00916 * - All fields must be valid (checked by internal_validate_config)
00917 * - Contents copied to handle (config can be stack-allocated)
00918 * - Sensor/motor arrays must remain valid (only pointers copied)
00919 *
00920 * @return Initialization result
00921 * @retval k_rx_ok System initialized successfully, ready to start
00922 * @retval k_rx_err_null_ptr handle, config, config->sensors/motors, or any sensor/motor
00923 *         handle in arrays is nullptr
00924 * @retval k_rx_err_invalid_arg sensor_count not in [1, 8]; motor_count not in [1, 4];
00925 *         threshold_cm not in [2.0, 400.0]; debounce_samples not in
00926 *         [1, 10]; poll_interval_ms not in [10, 1000]
00927 * @retval k_rx_err_invalid_state handle->initialized is already true
00928 * @retval k_rx_err_rtos_error ThreadX event flags or task creation failed
00929 *
00930 * @pre handle must point to valid rx_obstacle_detect_t storage
00931 * @pre config must point to valid configuration
00932 * @pre All sensors in config must be initialized via rx_hcsr04_init()
00933 * @pre All motors in config must be initialized via rx_motor_init()
00934 * @pre ThreadX must be running (tx_kernel_enter() completed)
00935 *
00936 * @post On success: handle->initialized = true
00937 * @post On success: ThreadX task created but suspended
00938 * @post On success: handle->state = k_obstacle_detect_state_stopped
00939 * @post On failure: handle unchanged (or partially cleared)
00940 * @post On failure: No ThreadX resources allocated
00941 *
00942 * @note Not thread-safe - do not call concurrently on same handle
00943 * @note Re-entrant for different handles
00944 * @note Task starts suspended - call rx_obstacle_detect_start() to begin
00945 *
00946 * @warning Do not modify handle fields after initialization
00947 * @warning Do not deinitialize sensors/motors while detection is running
00948 * @warning Ensure ThreadX kernel is running before calling
00949 *
00950 * @par Performance:
00951 * Execution time: ~100us (ThreadX task creation overhead)
00952 * Memory: 2.6 KB static (handle structure with embedded stack)
00953 *
00954 * @par Example:
00955 * @code{.c}
00956 * // Declare handle (static or global recommended for large struct)
00957 * static rx_obstacle_detect_t detector;
00958 *
00959 * // Setup sensor and motor arrays
00960 * rx_hcsr04_t* sensors[] = {&sensor_front, &sensor_rear};
00961 * rx_motor_handle_t* motors[] = {&motor_left, &motor_right};
00962 *
00963 * // Configure detection system
00964 * rx_obstacle_detect_config_t config = {
00965 *     .sensors = sensors,
00966 *     .sensor_count = 2,
00967 *     .motors = motors,
00968 *     .motor_count = 2,
00969 *     .detection_threshold_cm = 30.0f,
00970 *     .debounce_samples = 3,
00971 *     .poll_interval_ms = 20,
00972 *     .callback = my_callback,
00973 *     .user_data = &my_context
00974 * };
00975 *
00976 * // Initialize
00977 * rx_err_t ret = rx_obstacle_detect_init(&detector, &config);
00978 * if (ret != k_rx_ok) {
00979 *     printf("Init failed: %d\n", ret);
00980 *     return ret;
00981 * }
00982 *

```

```

00983 * // Start monitoring
00984 * ret = rx_obstacle_detect_start(&detector);
00985 * @endcode
00986 *
00987 * @see rx_obstacle_detect_config_t Configuration structure
00988 * @see rx_obstacle_detect_start() Start monitoring after init
00989 * @see rx_obstacle_detect_deinit() Cleanup and release resources
00990 * @see internal_validate_config() Internal configuration validation
00991 *
00992 * @since Version 1.0.0
00993 *
00994 * @par NASA Power of 10 Compliance:
00995 * - Rule 3: [OK] No dynamic memory - all allocation is static
00996 * - Rule 5: [OK] Minimum 6 preconditions validated
00997 * - Rule 7: [OK] All ThreadX return values checked
00998 */
00999 [[nodiscard]] rx_err_t rx_obstacle_detect_init(rx_obstacle_detect_t* handle,
10000                                         const rx_obstacle_detect_config_t* config);
10001
10002 /**
10003 * @brief Deinitialize obstacle detection system
10004 *
10005 * Stops detection task, releases ThreadX resources, and resets handle state.
10006 * Does NOT deinitialize the sensor or motor handles (caller's responsibility).
10007 *
10008 * @param[in,out] handle Obstacle detection handle
10009 *
10010 * @return k_rx_ok on success
10011 * @return k_rx_err_null_ptr if handle is nullptr
10012 * @return k_rx_err_invalid_state if not initialized
10013 */
10014 [[nodiscard]] rx_err_t rx_obstacle_detect_deinit(rx_obstacle_detect_t* handle);
10015
10016 /*
=====
10017 * Public API - Control
10018 *
=====
*/
10019
10020 /**
10021 * @brief Start obstacle detection monitoring
10022 *
10023 * Starts the detection task to begin polling sensors. If an obstacle is
10024 * already detected, motors will be stopped immediately.
10025 *
10026 * @param[in,out] handle Obstacle detection handle
10027 *
10028 * @return k_rx_ok on success
10029 * @return k_rx_err_null_ptr if handle is nullptr
10030 * @return k_rx_err_invalid_state if not initialized
10031 */
10032 [[nodiscard]] rx_err_t rx_obstacle_detect_start(rx_obstacle_detect_t* handle);
10033
10034 /**
10035 * @brief Stop obstacle detection monitoring
10036 *
10037 * Stops the detection task. Motors will NOT be automatically restarted even
10038 * if they were previously stopped due to obstacle detection.
10039 *
10040 * @param[in,out] handle Obstacle detection handle
10041 *
10042 * @return k_rx_ok on success
10043 * @return k_rx_err_null_ptr if handle is nullptr
10044 * @return k_rx_err_invalid_state if not initialized
10045 */
10046 [[nodiscard]] rx_err_t rx_obstacle_detect_stop(rx_obstacle_detect_t* handle);
10047
10048 /**
10049 * @brief Resume motors after obstacle cleared
10050 *
10051 * Clears the obstacle state and allows motors to be controlled again.
10052 * This is a manual override - the user must verify the obstacle is truly
10053 * cleared before calling this function.
10054 *
10055 * If detection is still running and obstacle is still present, motors
10056 * will be stopped again on the next poll cycle.
10057 *
10058 * @param[in,out] handle Obstacle detection handle
10059 *
10060 * @return k_rx_ok on success
10061 * @return k_rx_err_null_ptr if handle is nullptr
10062 * @return k_rx_err_invalid_state if not initialized
10063 */
10064 [[nodiscard]] rx_err_t rx_obstacle_detect_clear_obstacle(rx_obstacle_detect_t* handle);
10065
10066 /*
10067 */

```

```

=====
01068 * Public API - Status
01069 *
=====
01070 */
01071
01072 /**
01073  * @brief Get current detection state
01074  *
01075  * Returns the current state of the obstacle detection system.
01076  *
01077  * @param[in] handle Obstacle detection handle
01078  * @param[out] out_state Current state
01079  *
01080  * @return k_rx_ok on success
01081  * @return k_rx_err_null_ptr if handle or out_state is nullptr
01082  * @return k_rx_err_invalid_state if not initialized
01083  */
01084 [[nodiscard]] rx_err_t rx_obstacle_detect_get_state(const rx_obstacle_detect_t* handle,
01085                                                     rx_obstacle_detect_state_t* out_state);
01086
01087 /**
01088  * @brief Check if obstacle is currently detected
01089  *
01090  * Returns true if any sensor currently detects an obstacle within the
01091  * threshold distance (after debouncing).
01092  *
01093  * @param[in] handle Obstacle detection handle
01094  *
01095  * @return true if obstacle detected
01096  * @return false if no obstacle or handle is nullptr/invalid
01097  */
01098 [[nodiscard]] bool rx_obstacle_detect_is_obstacle_detected(const rx_obstacle_detect_t* handle);
01099
01100 /**
01101  * @brief Get obstacle detection statistics
01102  *
01103  * Returns statistics about detection performance and events.
01104  *
01105  * @param[in] handle Obstacle detection handle
01106  * @param[out] out_total_polls Total sensor polls (can be NULL)
01107  * @param[out] out_obstacle_events Total obstacle events (can be NULL)
01108  * @param[out] out_false_positives Debounced false positives (can be NULL)
01109  *
01110  * @return k_rx_ok on success
01111  * @return k_rx_err_null_ptr if handle is nullptr
01112  * @return k_rx_err_invalid_state if not initialized
01113  */
01114 [[nodiscard]] rx_err_t rx_obstacle_detect_get_stats(const rx_obstacle_detect_t* handle,
01115                                                     uint32_t* out_total_polls,
01116                                                     uint32_t* out_obstacle_events,
01117                                                     uint32_t* out_false_positives);
01118
01119 /**
01120  * @brief Reset statistics counters
01121  *
01122  * Resets all statistics counters to zero.
01123  *
01124  * @param[in,out] handle Obstacle detection handle
01125  *
01126  * @return k_rx_ok on success
01127  * @return k_rx_err_null_ptr if handle is nullptr
01128  * @return k_rx_err_invalid_state if not initialized
01129  */
01130 [[nodiscard]] rx_err_t rx_obstacle_detect_reset_stats(rx_obstacle_detect_t* handle);
01131
01132 */
=====
01133 * Internal Functions Exposed for Unit Testing
01134 *
=====
01135 */
01136
01137 #ifdef UNIT_TEST
01138 #include "rx_check.h"
01139
01140 /**
01141  * @brief Poll all sensors and update obstacle state (exposed for unit testing)
01142  * @param[in,out] handle Obstacle detection handle
01143  *
01144  * @return k_rx_ok on success
01145  */
01146 rx_err_t internal_poll_sensors(rx_obstacle_detect_t* handle);
01147
01148 /**
01149  * @brief Stop all configured motors (exposed for unit testing)
01150  * @param[in] handle Obstacle detection handle
01151  *
01152  * @return k_rx_ok if all motors stopped
01153  */

```

```

01151 */
01152 rx_err_t internal_stop_all_motors(const rx_obstacle_detect_t* handle);
01153
01154 /**
01155  * @brief Invoke user callback if registered (exposed for unit testing)
01156  * @param[in] handle Obstacle detection handle
01157  * @param[in] obstacle_detected True if obstacle detected, false if cleared
01158  * @param[in] sensor_idx Sensor index that triggered the event
01159  * @param[in] distance_cm Measured distance in centimeters
01160  */
01161 void internal_invoke_callback(const rx_obstacle_detect_t* handle,
01162                             bool obstacle_detected,
01163                             uint8_t sensor_idx,
01164                             float distance_cm);
01165
01166 /**
01167  * @brief Control the outer task loop in unit test builds
01168  *
01169  * @details
01170  * Sets the s_task_running flag that controls the outer while loop of
01171  * internal_detection_task_entry(). Set to false before calling the
01172  * task entry to make it execute exactly one outer iteration and then exit.
01173  *
01174  * @param[in] running True to keep the loop running, false to stop after one iteration
01175  */
01176 void rx_obstacle_detect_test_set_task_running(bool running);
01177
01178 /**
01179  * @brief Exercise internal_detection_task_entry for unit test coverage
01180  *
01181  * @details
01182  * Calls internal_detection_task_entry with the provided handle cast to ULONG.
01183  * Exposed only in UNIT_TEST builds via this header.
01184  *
01185  * @param[in] input Handle cast to ULONG as the task entry expects
01186  */
01187 void rx_obstacle_detect_test_run_task_entry(ULONG input);
01188 #endif /* UNIT_TEST */
01189
01190 #ifdef __cplusplus
01191 }
01192 #endif

```

4.3 libs/rx_obstacle_detect/src/rx_obstacle_detect.c File Reference

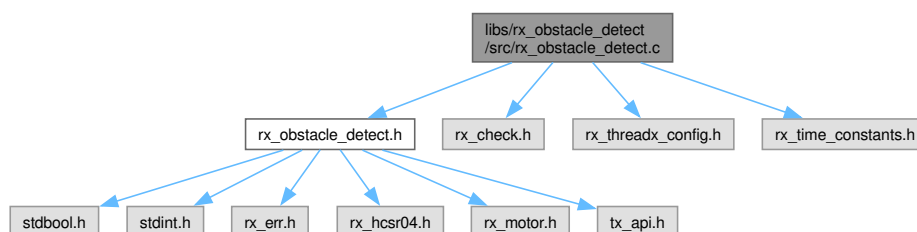
Safety-Critical Obstacle Detection Implementation with ThreadX Task.

```

#include "rx_obstacle_detect.h"
#include "rx_check.h"
#include "rx_threadx_config.h"
#include "rx_time_constants.h"

```

Include dependency graph for rx_obstacle_detect.c:



Macros

- #define [TASK_SHOULD_CONTINUE\(\)](#)
In production builds the task runs forever.

Enumerations

- enum [event_flags_t](#) : uint8_t { [k_event_flag_start](#) = 0x01 , [k_event_flag_stop](#) = 0x02 }
ThreadX event flag bit definitions for detection task control.
- enum [validation_constants_t](#) : uint16_t {
[k_min_threshold_cm](#) = 2 , [k_max_threshold_cm](#) = 400 , [k_min_poll_interval](#) = 10 ,
[k_max_poll_interval](#) = 1000 ,
[k_min_debounce](#) = 1 , [k_max_debounce](#) = 10 }
Configuration parameter validation limits.
- enum [task_constants_t](#) : uint8_t { [k_min_sleep_ticks](#) = 1 }

Functions

- static void [internal_detection_task_entry](#) (ULONG input)
- static rx_err_t [internal_validate_config](#) (const [rx_obstacle_detect_config_t](#) *config)
Validate obstacle detect configuration.
- static void [internal_copy_config](#) ([rx_obstacle_detect_t](#) *handle, const [rx_obstacle_detect_config_t](#) *config)
Copy configuration fields and handles into the detection handle.
- static rx_err_t [internal_create_threadx_resources](#) ([rx_obstacle_detect_t](#) *handle)
Create ThreadX event flags and detection thread.
- static void [internal_run_detection_loop](#) ([rx_obstacle_detect_t](#) *const handle, const ULONG sleep_ticks)
Inner detection loop that polls sensors until stopped or error.
- static void [internal_process_sensor_reading](#) ([rx_obstacle_detect_t](#) *const handle, const uint8_t sensor_idx, const float distance_cm)
Process a single sensor distance reading through debounce logic.
- static void [internal_update_obstacle_state](#) ([rx_obstacle_detect_t](#) *const handle, rx_err_t *const out_err)
Evaluate aggregate obstacle state across all sensors and control motors.
- RX_STATIC_TESTABLE rx_err_t [internal_stop_all_motors](#) (const [rx_obstacle_detect_t](#) *handle)
Stop all configured motors.
- RX_STATIC_TESTABLE rx_err_t [internal_poll_sensors](#) ([rx_obstacle_detect_t](#) *handle)
Poll all sensors and update obstacle state.
- RX_STATIC_TESTABLE void [internal_invoke_callback](#) (const [rx_obstacle_detect_t](#) *handle, bool obstacle_detected, uint8_t sensor_idx, float distance_cm)
- rx_err_t [rx_obstacle_detect_init](#) ([rx_obstacle_detect_t](#) *handle, const [rx_obstacle_detect_config_t](#) *config)
Initialize obstacle detection system with sensors and motors.
- rx_err_t [rx_obstacle_detect_deinit](#) ([rx_obstacle_detect_t](#) *handle)
Deinitialize obstacle detection system and release all resources.
- rx_err_t [rx_obstacle_detect_start](#) ([rx_obstacle_detect_t](#) *handle)
Start obstacle detection monitoring.
- rx_err_t [rx_obstacle_detect_stop](#) ([rx_obstacle_detect_t](#) *handle)
Stop obstacle detection monitoring.
- rx_err_t [rx_obstacle_detect_clear_obstacle](#) ([rx_obstacle_detect_t](#) *handle)
Clear active obstacle state and reset per-sensor debounce counters.
- rx_err_t [rx_obstacle_detect_get_state](#) (const [rx_obstacle_detect_t](#) *handle, [rx_obstacle_detect_state_t](#) *out_state)
Get the current obstacle detection state.
- bool [rx_obstacle_detect_is_obstacle_detected](#) (const [rx_obstacle_detect_t](#) *handle)

Check whether an obstacle is currently detected (boolean convenience).

- `rx_err_t rx_obstacle_detect_get_stats` (const `rx_obstacle_detect_t` *handle, `uint32_t` *out_total_polls, `uint32_t` *out_obstacle_events, `uint32_t` *out_false_positives)

Retrieve accumulated diagnostic statistics from obstacle detection system.

- `rx_err_t rx_obstacle_detect_reset_stats` (`rx_obstacle_detect_t` *handle)

Reset all accumulated diagnostic statistics counters to zero.

Variables

- static const `rx_obstacle_detect_t s_zero_handle` = {}
Zero-initialized obstacle detect template for stack-safe initialization.

4.3.1 Detailed Description

Safety-Critical Obstacle Detection Implementation with ThreadX Task.

Implements the obstacle detection system using a dedicated ThreadX task that autonomously polls HC-SR04 ultrasonic sensors and triggers emergency motor stops when obstacles breach the configured safety threshold. This module provides collision avoidance through continuous sensor monitoring and immediate emergency stop capability.

Implementation Architecture:

The system operates through three main components:

1. Detection Task ([internal_detection_task_entry](#)): ThreadX task running at configured priority
2. Polling Loop ([internal_poll_sensors](#)): Iterates through all sensors, applies debouncing
3. Emergency Stop ([internal_stop_all_motors](#)): Halts all motors when obstacle confirmed

Task State Machine: